3 Angle Modulation

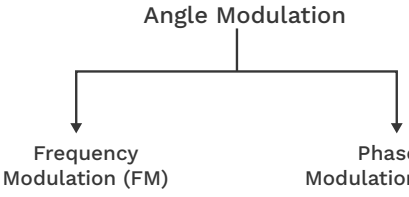
Objective

After covering this chapter, user will be able to understand:

1. What is angle modulation
2. Frequency and phase modulation
3. Relation between PM and FM
4. Generation of FM and PM
5. Demodulation of FM and PM
6. Carson's rule

Introduction

- There is another method of modulating a sinusoidal carrier wave, namely, angle modulation in which either the phase or frequency of the carrier wave is varied according to the message signal.
- In this method of modulation, the amplitude of the carrier wave is maintained constant.



- Frequency of the carrier is varied according to the message signal
- Phase angle of the carrier is varied according to the message signal

Some Basic Fundamentals

Signal = x(t)	$\lvert x(t)\rvert_{max}$
$A\cos\omega_0 t \longrightarrow$	A
$-A\cos\omega_0 t \longrightarrow$	A
$A\cos\omega_0 t + B\sin\omega_0 t \longrightarrow$	$\sqrt{A^2+B^2}$
$A\cos\omega_0 t + B\cos\omega_0 t \longrightarrow$	$\lvert A+B\rvert$
$A\sin\omega_0 t + B\sin\omega_0 t \longrightarrow$	$\lvert A+B\rvert$

$x(t)$	$\lvert x(t)_{max}\rvert$
$\cos\omega_1 t + B\cos\omega_2 t \longrightarrow$	$\lvert A+B\rvert$
$A\sin\omega_1 t + B\sin\omega_2 t \longrightarrow$	$\lvert A+B\rvert$
$A\cos\omega_1 + B\sin\omega_2 t \longrightarrow$	$\rightarrow \nearrow \lvert A+B\rvert$ if $A=B$ $\searrow < \lvert A+B\rvert$ if $A\neq B$

Example: $x(t) = A\sin\omega_0 t + B\sin(\omega_0 t + \theta)$

$\lvert x(t)\rvert_{max} = ?$

Solution:

$$x(t) = (A + B\cos\theta)\sin\omega_0 t + (B\sin\theta)\cos\omega_0 t$$

$$\lvert x(t)\rvert_{max} = \sqrt{(A+B\cos\theta)^2 + (B\sin\theta)^2}$$

$$= \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

Example:

$$x(t) = 12\cos(2\pi \times 10^6 t) + 5\sin(2\pi \times 10^6 t)$$

$\lvert x(t)\rvert_{max} = ?$

Solution:

$$\lvert x(t)\rvert_{max} = \sqrt{12^2 + 5^2} = 13$$

Concept of Instantaneous Angle and Instantaneous Frequency

$$S(t) = A_c\cos(\theta_i(t))$$

$\theta_i(t)$: Instantaneous angle: (rad)

$\frac{d\theta_i(t)}{dt} = \omega_i(t)$:

Instantaneous angular frequency

$f_i(t) = \frac{1}{2\pi}\frac{d\theta_i(t)}{dt} \rightarrow$

Instantaneous liners frequency

$\theta_i(t)$: Given

$f_i(t)$: Given

$$\rightarrow \frac{d\theta_i(t)}{dt} = \omega_i(t)$$

$$\rightarrow \omega_i(t) = 2\pi f_i(t)$$

$\rightarrow$ Running Intergrator

$$\rightarrow f_i(t) = \frac{\omega_i(t)}{2\pi}$$

$$\rightarrow \theta_i(t) = \int_{-\infty}^{t} \omega_i(t)dt$$

$\downarrow$

$$\theta_i(t) = \int_{-\infty}^{t} \omega_i(t)dt$$

Example:

$$s(t) = 2\cos\left(20\pi t + \pi t^2\right)$$

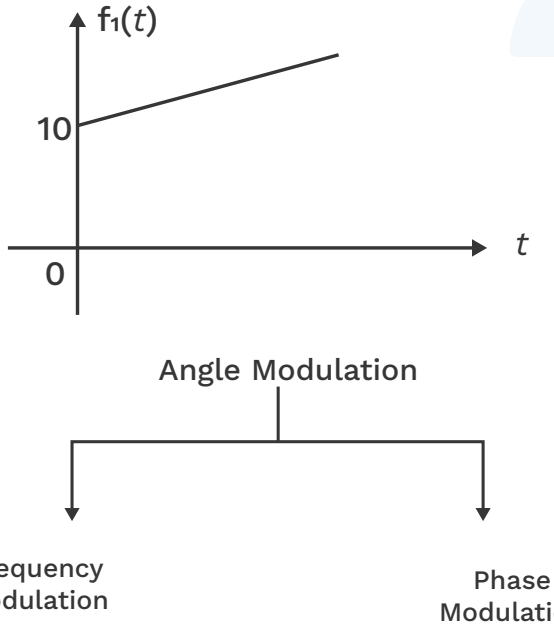
Calculate instantaneous frerquency.

Solution:

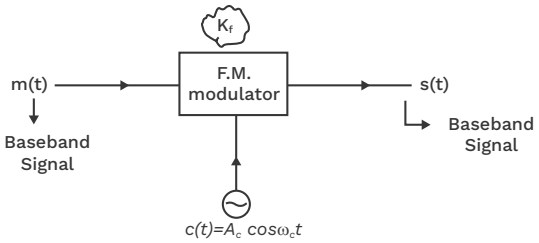
$$\theta_i(t) = 20\pi t + \pi t^2$$

$$\omega_i(t) = \frac{d\theta_i(t)}{dt} = 20\pi + 2\pi t$$

$$f_i(t) = 10 + t$$



Frequency modulation as angle modulation:



$m(t)$: Baseband message signal

$c(t)$: Carrier before modulation = $A_c \cos \omega_c t$

General expression of angle modulated signal:

$$S_{angle}(t) = A_c \cos\left(\omega_c t + \Delta\phi(t)\right)$$

Instantaneous angle

$$\theta_i(t) = \omega_c t + \Delta\phi(t)$$

$$\omega_i(t) = \frac{d\theta_i(t)}{dt} = \omega_c + \frac{d\Delta\phi(t)}{dt}$$

If angle modulation is FM

$$\frac{d\Delta\phi(t)}{dt} \ \alpha \ m(t)$$

$$\frac{d\Delta\phi(t)}{dt} = K_f m(t)$$

K_f : Frequency sensitivity of frequency modulator

$$\omega_i(t) = \omega_c + K_f m(t)$$

Instantaneous frequency varies linearly with m(t)

$\downarrow$

$\theta_i(t)$ varies with

$$m(t) \Rightarrow \boxed{\theta_i(t) = \omega_c t + \int_{-\infty}^{t} K_f m(t)dt}$$

$\downarrow$

Frequency Modulation

Frequency Modulation

1. It is a type of angle modulation.
2. The angle of modulated signal varies with baseband message signal so that baseband message signal can be converted into bandpass signal
3. The angle of modulated signal varies with *m(t)* because instantaneous frequency varies linearly with *m(t)*.

$$c(t) = A\cos \omega_c t \begin{matrix} \nearrow \omega_c \\ \searrow 0^0 \end{matrix}$$

$$S_{angle} = A_c \cos\left(\omega_c t + \Delta\phi(t)\right)$$

$\Delta\phi(t)$: Phase deviation

$$\theta_i(t) = \omega_c t + \Delta\phi(t)$$

$$\omega_i(t) = \omega_c + \frac{d\Delta\phi(t)}{dt}$$

$$\boxed{\begin{matrix} A_{angular} \\ \Delta\omega(t) : \text{Frequency deviation} \end{matrix}}$$

$$\omega_i(t) = \omega_c + \Delta\omega(t)$$

$$\omega_i(t) = \omega_c + \frac{d\Delta\phi(t)}{dt}$$

$$\boxed{\frac{d\Delta\phi(t)}{dt} = \Delta\omega(t) = K_f m(t)}$$

4. In frequency modulation, angular (linear) frequency deviation varies linearly with m(t).

Basic terms:

1. $c(t) = A_c \cos\left(\omega_c t\right) \begin{matrix} \nearrow \text{Angle} = \omega_c t \\ \searrow \text{Frequency} = \omega_c \end{matrix}$
2. $S_{angle}(t) = A_c \cos\left(\omega_c t + \Delta\phi(t)\right) \begin{matrix} \nearrow \text{Angle} = ? \\ \searrow \text{Frequency} = ? \end{matrix}$

   $$\theta_i(t) = \omega_c t + \Delta\phi(t)$$
3. $\theta_i(t)$: Instantaneous angle
4. $\Delta\phi(t)$ = Instantaneous phase deviation
5. Maximum instantaneous phase deviation $= \left|\Delta\phi(t)\right|_{max}$

$$\frac{d\theta_i(t)}{dt} = \omega_c + \frac{d\Delta\phi(t)}{dt}$$

$$\boxed{\omega_i(t) = \omega_c + \Delta\omega(t)}$$

$\omega_i(t) \rightarrow$ Instantaneous angular frequency

$\Delta\omega(t) \rightarrow$ Instantaneous frequency deviation

Maximum instantaneous angular frequency deviation $= \left|\Delta\omega(t)\right|_{max}$

Relation between $\Delta\omega(t)$ and $\Delta\phi(t)$:

$$\theta_i(t) = \omega_c t + \Delta\phi(t)$$

$$\omega_i(t) = \omega_c + \frac{d\Delta\phi(t)}{dt} = \omega_c + \Delta\omega(t)$$

$$\boxed{\Delta\omega(t) = \frac{d\Delta\phi(t)}{dt}}$$

$$\boxed{\Delta\phi(t) = \int_{-\infty}^{t} \Delta\omega(t)dt}$$

Mathematical definition of FM:

Case 1: $K_f = \frac{rad}{V - \sec}$

FM defination $\omega_i(t) = \omega_c + K_f m(t)$

Important frequency calculation: $K_f : \frac{rad}{V - \sec}$

$$\omega_i(t) = \omega_c + K_f m(t)$$

Instantaneous frequency deviation

$$= \Delta\omega(t) = K_f m(t)$$

Frequency deviation in positive direction:

$$\left\{\Delta\omega(t)\right\}_{max} = K_f \left\{m(t)\right\}_{max}$$

Frequency deviation in negative direction:

$$\left\{\Delta\omega(t)\right\}_{min} = K_f \left\{m(t)\right\}_{min}$$

Maximum value of instantaneous frequency:

$$\left\{\omega_i(t)\right\}_{max} = \omega_c + (\Delta\omega(t))_{max}$$

Minimum value of inst. frequency:

$$\left\{\omega_i(t)\right\}_{min} = \omega_c + (\Delta\omega(t))_{min}$$

Peak to peak frequency deviation:

$$(\Delta\omega)_{P-P} = \left\{\omega_i(t)\right\}_{max} - \left\{\omega_i(t)\right\}_{min}$$

Maximum frequency deviation $= \left|\Delta\omega(t)\right|_{max}$

Modulation index or deviation ratio of FM

$$\beta_{FM} = \frac{\left|\Delta\omega(t)\right|_{max}}{\omega_{max}}$$

$$= \frac{\text{maximum frequency deviation}}{\text{max frequency component of } m(t)}$$

$$\beta_{FM} = \frac{K\left|m(t)\right|_{max}}{\omega_{max}}$$

Important phase calculation: $K_f \frac{(rad)}{V \sec}$

FM $\omega_i(t) = \omega_c + K_f m(t)$

$$\theta_i(t) = \int_{-\infty}^{t} \omega_i(\tau) d\tau$$

$$\theta_i(t) = \omega_c t + \int_{-\infty}^{t} K_f m(\tau) d\tau$$

Instantaneous phase deviation in FM:

$$\Delta\phi(t) = K_f \int_{-\infty}^{t} m(\tau) d\tau$$

Maximum phase deviation in FM:

$$\Delta\phi(t)\big|_{\max} = K_f \left| \int_{-\infty}^{t} m(\tau) d\tau \right|$$

General expression of FM:

$$K_f : \frac{rad}{V - \sec}$$

$$\omega_i(t) = \omega_c + K_f m(t)$$

$$\text{Sangle } (t) = A_c \cos(\omega_c t + \Delta\phi(t))$$

$$= A_c \cos(\theta_i(t))$$

$$\text{Sangle } (t) = A_c \cos(\omega_c t + \int_{-\infty}^{t} K_f m(\tau) d\tau)$$

Case 2:

Unit of K_f : $Hz / Volt$

FM defination :$\rightarrow f_i(t) = f_c + K_f m(t)$

Frequency calculation in FM: $K_f(Hz / Volt)$

$$f_i(t) = f_c + K_f m(t)$$

1. Instantaneous frequency deviation
   $= \Delta f(t) = K_f m(t) Hz$
2. Instantaneous frequency deviation in +ve direction
   $= \{\Delta f(t)\}_{\max} = K_f \{m(t)\}_{\max}$
3. Instantaneous frequency deviation in -ve direction
   $= \{\Delta f(t)\}_{\min} = K_f \{m(t)\}_{\min}$
4. Maximum instantaneous frequency:
   $\{f_i(t)\}_{\max} = f_c + \{\Delta f(t)\}_{\max}$
5. Minimum instantaneous frequency:
   $\{f_i(t)\}_{\min} = f_c + \{\Delta f(t)\}_{\min}$
6. Peak to peak frequency deviation:
   $\Delta f_{p-p} = \{f(t)\}_{\max} - \{f(t)\}_{\min}$
7. Maximum frequency deviation
   $|\Delta f(t)|_{\max} = K_f |m(t)|_{\max}$
8. Deviation ratio:
   $$\beta_{FM} = D = \frac{|\Delta f(t)|_{\max}}{f_{\max}} = \frac{K_f |m(t)|_{\max}}{f_{\max}}$$

Important phase calculation:

$K_f(Hz / Volt)$

$F.M. \rightarrow f_i(t) = f_c + K_f m(t)$

$$\omega_i(t) = 2\pi f_c + 2\pi K_f m(t) = \omega_c + 2\pi K_f m(t)$$

$$\theta_i(t) = \int_{-\infty}^{t} \omega_i(\tau) d\tau = \omega_c t + 2\pi K_f \int_{-\infty}^{t} m(\tau) d\tau$$

$$\theta_i(t) = \omega_c t + 2\pi K_f \int_{-\infty}^{t} m(\tau) d\tau$$

1. Instantaneous phase deviation:
   $$\Delta\phi(t) = 2\pi K_f \int_{-\infty}^{t} m(\tau) d\tau$$
2. Maximum phase deviation:
   $$|\Delta\phi(t)|_{\max} = 2\pi K_f \left| \int_{-\infty}^{t} m(\tau) d\tau \right|_{\max}$$

General expression of FM:

$K_f = Hz / Volt$

$$\text{Sangle } (t) = A_c \cos(\theta_i(t))$$

$$F.M. \quad f_i(t) = f_c + K_f m(t)$$

$$\theta_i(t) = \omega_c t + 2\pi K_f \int_{-\infty}^{t} m(\tau) d\tau$$

$$S_{FM}(t) = A_c \cos(\omega_c t + 2\pi K_f \int_{-\infty}^{t} m(\tau) d\tau)$$

Conclusion

$$K_f : \frac{rad}{V-\sec} \longrightarrow \omega_i(t) = \omega_c + K_f m(t) \longrightarrow$$

or

$$f_i(t) = f_c + \frac{K_f}{2\pi} m(t)$$

$$K_f : \frac{Hz}{Volt} \longrightarrow f_i(t) = f_c + K_f m(t) \longrightarrow$$

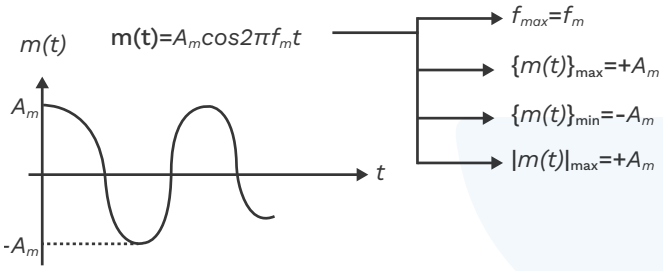
or

$$\omega_i(t) = \omega_c + 2\pi K_f m(t)$$

Special case

Case 1:

$$m(t) = A_m \cos 2\pi f_m t$$



$$\rightarrow K_f : \frac{Hz}{volt}$$

$$\rightarrow F.M : f_i(t) = (f_c + Kf_m(t))$$

Frequency related calculation:

1. $\Delta f(t) = K_f m(t) = K_f A_m \cos 2\pi f_m t \ \ Hz$
2. $\{\Delta f(t)\}_{\max} = K_f A_m \ \ Hz$
3. $\{\Delta f(t)\}_{\min} = -K_f A_m \ Hz$
4. $\{f_i(t)\}_{\max} = (f_c + K_f A_m) \ Hz$
5. $\{f_i(t)\}_{\min} = (f_c - K_f A_m) \ Hz$
6. $\Delta f_{p-p} = 2K_p A_m \ Hz$
6. $|\Delta f(t)|_{\max} = K_f |m(t)|_{\max} = K_f A_m = \frac{\Delta f_{p-p}}{2}$
7. $\beta_{FM} = \frac{|\Delta f(t)|_{\max}}{f_{\max}} = \frac{K_f A_m}{f_m}$

Phase calculation:

$K_f : Hz / Volt \ \ m(t) = A_m \cos \omega_m t$

FM.: $f_i(t) = f_c + K_f m(t)$

$$\theta_i(t) = \omega_c t + \frac{2\pi K_f}{2\pi f_m}[A_m \sin 2\pi f_m t]$$

$$\theta_i(t) = \omega_c t + \frac{K_f A_m}{f_m} \sin 2\pi f_m t$$

$$\Delta\phi(t) = \frac{K_f A_m}{f_m} \sin 2\pi f_m t$$

$$|\Delta\phi(t)|_{\max} = \frac{K_f A_m}{f_m} = \beta$$

General expression:

$$S_{FM}(t) = A_c \cos(\theta_i(t))$$

$$\theta_i(t) = \omega_c t + \beta_{FM} \sin(2\pi f_m t)$$

$$S_{FM}(t) = A_c \cos(\omega_c t + \beta_{FM} \sin(2\pi f_m t))$$

Points to Remember

$$\rightarrow K_f(Hz / Volt)$$

$$\rightarrow m(t) = A_m \cos 2\pi f_m t$$

1. $|\Delta f(t)|_{\max} = K_f A_m = \Delta f$

$$\frac{\Delta f_1}{\Delta f_2} = \frac{A_{m_1}}{A_{m_2}}$$

2. $\beta = \frac{\Delta f}{f_{\max}} = \frac{K_f A_m}{f_m}$

$$\frac{\beta_1}{\beta_2} = \frac{(A_{m_1} / f_{m_1})}{(A_{m_2} / f_{m_2})}$$

3. $|\Delta\phi(t)|_{\max} = \Delta\phi = \beta \ \ \frac{\Delta\phi_1}{\Delta\phi_2} = \frac{\beta_1}{\beta_2}$

Case 2: $m(t) = A_{m_1} \cos 2\pi f_{m_1} t + A_{m_2} \cos 2\pi f_{m_2} t$

$f_{\max} = (f_{m_1}, f_{m_2})_{\max}$

$K_f : Hz / Volt$

FM $\rightarrow f_i(t) = f_c + K_f m(t)$

Important frequency calculation:

$f_i(t) = f_c + K_f m(t)$

1. $\Delta f(t) = K_f m(t)$
$= K_f A_{m_1} \cos 2\pi f_{m_1} t + K_f A_{m_2} \cos 2\pi f_{m_2} t$

2. $\{\Delta f(t)\}_{max}$ =
3. $\{\Delta f(t)\}_{min}$ = $\Longrightarrow$ $\frac{d\Delta f(t)}{dt}=0 \longrightarrow$ maxima/minima
4. $\{f_i(t)\}_{max} = f_c + \{\Delta f(t)\}_{max}$
5. $\{f_i(t)\}_{min} = f_c + \{\Delta f(t)\}_{min}$
6. $\Delta f_{p-p} = \{f_i(t)\}_{max} - \{f_i(t)\}_{min}$
7. $|\Delta f(t)|_{max} = K_f A_{m_1} + K_f A_{m_2}$
8. $\beta_{FM} = D = \frac{|\Delta f(t)|_{max}}{f_{max}} = \frac{K_f A_{m_1} + K_f A_{m_2}}{f_{m_2}}$ If $f_{m_2} > f_{m_1}$

$$\beta_{FM} = \frac{K_f A_{m_1}}{f_{m_2}} + \frac{K_f A_{m_2}}{f_{m_2}} \longrightarrow B_1 + B_2$$

Important phase calculation:

K_f : Hz / $Volt$

$$f_i(t) = f_c + K_f m(t)$$

$$\omega_i(t) = \omega_c + 2\pi K_f m(t)$$

$$\theta_i(t) = \int_{-\infty}^{t} \omega_i(\tau) d\tau = \omega_c t + 2\pi K_f \int_{-\infty}^{t} m(\varepsilon) d\varepsilon$$

$$\theta_i(t) = \omega_c t + 2\pi K_f$$

$$\left[\frac{A_{m_1}}{2\pi f_{m_1}} \sin 2\pi f_{m_1} t + \frac{A_{m_2}}{2\pi f_{m_2}} \sin 2\pi f_{m_2} t\right]$$

$$\theta_i(t) = \omega_c t + \frac{K_f A_{m_1}}{f_{m_1}} \sin 2\pi f_{m_1} t + \frac{K_f A_{m_2}}{f_{m_2}} \sin 2\pi f_{m_2} t$$

$$\Delta\phi(t) = \frac{K_f A_{m_1}}{f_{m_1}} \sin 2\pi f_{m_1} t + \frac{K_f A_{m_2}}{f_{m_2}} \sin 2\pi f_{m_2} t$$

$$|\Delta\phi(t)|_{max} = \underset{\downarrow}{\frac{K_f A_{m_1}}{f_{m_1}}} + \underset{\downarrow}{\frac{K_f A_{m_2}}{f_{m_2}}}$$
$$\beta_1 \qquad \beta_2$$

General expression:

$$S_{FM}(t) = A_c \cos\left(\theta_i(t)\right)$$

$$S_{FM}(t) = A_c \cos\left[\omega_c t + \beta_1 \sin 2\pi f_{m_1} t + \beta_2 \sin 2\pi f_{m_2} t\right]$$

$$\beta_1 = \frac{K_f A_{m_1}}{f_{m_1}} \quad \beta_2 = \frac{K_f A_{m_2}}{f_{m_2}}$$

Conclusion

If general expression of FM signal for multitone m(t) is given

$$S_{FM}(t) =$$

$$A_c \cos\left[2\pi f_c t + \beta_1 \sin 2\pi f_{m_1} t + \beta_2 \sin 2\pi f_{m_2} t\right]$$

$$f_{max} = \max\left(f_{m_1}, f_{m_2}\right)$$

Phase Modulation

$m(t)$:→ 'Baseband message signal'

$c(t)$ → Carrier before modulation

$$S_{angle}(t) = A_c \cos\left(\omega_c t + \Delta\phi(t)\right)$$

If $\Delta\phi(t) \propto m(t)$

$$\Delta\phi(t) = K_p m(t)$$
$$rad \qquad volt$$

K_p : Phase sensitivity of phase modulaion

$$K_p = \frac{\text{rad}}{\text{volt}}$$

Def^n :

1. It is an angle modulation.
2. In this angle of the modulated signal varies with baseband message signal in order to convert it into bandpass signal.
3. The angle of modulated signal varies with $m(t)$ because $\Delta\phi(t)$ varies linearly with $m(t)$.

$\Delta\omega(t) \alpha m(t) \rightarrow F.M.$
$\Delta\phi(t) \alpha m(t) \rightarrow P.M.$

Mathematical Def^n :

$$K_p : \frac{rad}{volt}$$

$$PM \rightarrow \theta_i(t) = \omega_c t + K_p m(t)$$

Important phase calculation:

$K_P : rad/volt$

$PM: \ \theta_i(t) = \omega_c t + K_p m(t):$

Instantaneous phase deviation

$= \Delta\phi(t) = K_p m(t)$

Maximum phase deviation

$= \left|\Delta\phi(t)\right|_{max} = K_p \left|m(t)_{max}\right|$

Important frequency calculation: $K_P \ rad/volt$

$PM: \ \theta_i(t) = \omega_c t + K_p m(t)$

$$\frac{d\theta_i C(t)}{dt} = \omega_i(t) = \omega_c + K_p \frac{dm(t)}{dt}$$

$$\omega_i(t) = \omega_c + \Delta\omega(t)$$

Instantaneous frequency deviation:

$$\Delta\omega(t) = K_p \frac{dm(t)}{dt}$$

Instantaneous frequency deviation in $+ve$ direction:

$$\left\{\Delta\omega(t)\right\}_{max} = K_p \left\{\frac{dm(t)}{dt}\right\}_{max}$$

Instantaneous frequency deviation in $-ve$ direction:

$$\left\{\Delta\omega(t)\right\}_{min} = K_p \left\{\frac{dm(t)}{dt}\right\}_{min}$$

Maximum instantaneous frequency:

$$\left\{\omega_i(t)\right\}_{max} = \omega_c + \left\{\Delta\omega(t)\right\}_{max}$$

Minimum instantaneous frequency:

$$\left\{\omega_i(t)\right\}_{min} = \omega_c + \left\{\Delta\omega(t)\right\}_{min}$$

Peak to peak frequency deviation:

$$\Delta\omega_{p.p} = \left\{\omega_i(t)\right\}_{max} - \left\{\omega_i(t)\right\}_{min}$$

Maximum frequency deviation:

$$\left|\Delta\omega(t)\right|_{max} = K_p \left|\frac{dm(t)}{dt}\right|_{max}$$

Modulation index or deviation ratio:

$$\beta_{PM} = \frac{\left|\Delta\omega(t)\right|_{max}}{\omega_{max}} = \frac{K_p \left|\frac{dm(t)}{dt}\right|_{max}}{\omega_{max}}$$

General expression of PM:

$$S_{PM}(t) = A_c \cos\left(\theta_i(t)\right)$$

$$S_{PM}(t) = A_c \cos\left(\omega_c t + K_p m(t)\right)$$

Special cases:

$$m(t) = A_m \cos 2\pi fmt$$

$$\Rightarrow PM: \theta_i(t) = \omega_c t + K_p \, m(t)$$

Phase calculation:

$$\rightarrow \Delta\phi(t) = K_p \, A_m \cos 2\pi f_m t$$

$$\left|\Delta\phi(t)\right|_{max} = K_p A_m$$

Frequency calculation:

$$K_p : \frac{rad}{Volt} \theta_i(t) = \omega_c t + K_p m(t)$$

$$\omega_i(t) = \omega_c + K_p \frac{dm(t)}{dt}$$

1. $\Delta\omega(t) = K_p \frac{dm(t)}{dt} = -K_p \, A_m \, \omega_m \sin \omega_m t$
2. $\left\{\Delta\omega(t)\right\}_{max} = K_p \, A_m \, \omega_m$
3. $\left\{\Delta\omega(t)\right\}_{min} = -K_p \, A_m \, \omega_m$
4. $\left\{\Delta\omega_i(t)\right\}_{max} = \omega_c + K_p \, A_m \, \omega_m$
5. $\left\{\Delta\omega_i(t)\right\}_{min} = \omega_c - K_p \, A_m \, \omega_m$
6. $(\Delta\omega)_{pp} = 2K_p \, A_m \, \omega_m$
7. $\left|\Delta\omega(t)\right|_{max} = K_p \left|\frac{dm(t)}{dt}\right|_{max} = K_p \, A_m \, \omega_m$
8. $\beta = \frac{\left|\Delta\omega(t)\right|_{max}}{\omega_{max}} = \frac{K_p \, A_m \, \omega_m}{\omega_m} = K_p A_m = \left|\Delta\phi(t)\right|_{max}$

General expression: $\rightarrow S_{PM}(t) = A_c \cos\left(\theta_i(t)\right)$

$$S_{PM}(t) = A_c \cos\left(\omega_c t + K_p A_m \cos 2\pi f_m t\right)$$

$$S_{PM}(t) = A_c \cos\left(\omega_c t + \beta_{PM} \cos 2\pi f_m t\right)$$

Conclusion

$$K_p : \frac{rad}{V} \quad m(t) = A_m \cos 2\pi f m_m t$$

1. $\left|\Delta\omega(t)\right|_{\max} = \Delta\omega = K_p A_m \omega_m$

   $$\frac{\Delta\omega_1}{\Delta\omega_2} = \frac{A_{m_1}\omega_1}{A_{m_2}\omega_2}$$

2. $\beta = K_p A_m \rightarrow \quad \frac{\beta_1}{\beta_2} = \frac{A_{m_1}}{A_{m_2}}$

3. $\left|\Delta\phi(t)\right|_{\max} = \Delta\phi = \beta$

Common Points to Avoid Mistakes

1. $S(t) = A_c \cos\left(\omega_c t + K_f \int_{-\infty}^{t} m(\tau) d\tau\right)$

   $$\rightarrow K_f : \frac{rad}{v - \sec}$$

2. $S(t) = A_c \cos\left(\omega_c t + 2\pi K_f \int_{-\infty}^{t} m(\tau) d\tau\right)$

   $$\rightarrow K_f : \frac{Hz}{volt}$$

3. $S(t) = A_c \cos\left(\omega_c t + K_p m(t)\right) \Rightarrow P.M.$

   $$\Rightarrow K_p : \frac{rad}{volt}$$

4. $m(t) = A_m \cos 2\pi f_m t \quad c(t) = A_c \cos 2\pi f_c t$

   $F.M.$ $S_{FM}(t) = A_c \cos\left(2\pi f_c t + \beta \sin 2\pi f_m t\right)$

   $$\beta = \frac{K_f A_m}{f_m}$$

   $P.M.$ $S_{FM}(t) = A_c \cos\left(2\pi f_c t + \beta \sin 2\pi f_m t\right)$

   $$\beta = K_p A_m$$

5. $m(t) = A_{m_1} \cos 2\pi f_{m_1} t + A_{m_2} \cos 2\pi f_{m_2} t$

   $$S_{FM}(t) = A_c \cos\begin{pmatrix} \omega_c t + \beta_1 \sin 2\pi f_{m_1} \\ t + \beta_2 \sin 2\pi f_{m_2} t \end{pmatrix}$$

   $$\beta_1 = \frac{K_f A_{m_1}}{f_{m_1}} \quad \beta_2 = \frac{K_f A_{m_2}}{f_{m_2}}$$

   $$S_{PM}(t) = A_c \cos\begin{pmatrix} \omega_c t + \beta_1 \cos 2\pi f_{m_1} t \\ +\beta_2 \cos 2\pi f_{m_2} t \end{pmatrix}$$

   $$\beta_1 = K_p A_{m_1} \quad \beta_2 = K_p A_{m_2}$$

6. $m(t) = A_m \cos 2\pi f_m t$

   $$F.M. - \begin{cases} \xrightarrow{k_f\, rad/v\sec} & \Delta\omega = K_f A_m, \Delta\phi = \beta \\ \xrightarrow[K_f : \frac{Hz}{Volt}]{} & \Delta f = K_f A_m \ , \ \beta \frac{K_f A_m}{f_m} \quad \Delta\phi = \beta \end{cases}$$

   $$PM \xrightarrow{K_p : rad/volt} \Delta\omega = K_p A_m \omega_m,$$

   $$\beta = K_p A_m \ \Delta\phi = \beta$$

Example: Calculate maximum phase deviation of following signals.

$$(i) \quad S(t) = A_c \cos\left(\omega_c t + \frac{t}{1+t^2}\right)$$

Solution: $S(t) = A_c \cos\left(\omega_c t + \Delta\phi(t)\right)$

$$\Delta\phi(t) = \frac{t}{1+t^2}$$

$$\frac{d\Delta\phi(t)}{dt} = 0 \rightarrow t = \pm 1$$

$$\Delta\phi(t = 1) = \frac{1}{2}$$

$$\Delta\phi(t = -1) = \frac{-1}{2}$$

$$\left|\Delta\phi(t)\right|_{\max} = \frac{1}{2}\ rad$$

2. $s(t) = A_c \cos\left(\omega_c t + \frac{4\sin 2t}{t}\right)$

Solution: $s(t) = A_c \cos\left(\omega_c t + \Delta\phi(t)\right)$

$$\left|\Delta\phi(t)\right| = \frac{4\sin 2t}{t} = 8\sin c\left(\frac{2t}{\pi}\right)$$

$$\left|\Delta\phi(t)\right|_{\max} = \theta_{rad} \qquad at\ \ t=0$$

Example: Expression of angle modulated is given by:

$$s(t) = \cos\begin{pmatrix} 4\pi \times 10^6 t + 60\pi \cos 150 t \\ +80\pi \sin 150 t \end{pmatrix}$$

1. Maximum phase deviation
2. Maximum frequency deviation
3. Deviation ratio

Solution: $s(t) = A_c \cos\left(\omega_c t + \Delta\phi(t)\right)$

I) $\Delta\phi(t) = 60\pi \cos 150t + 80\pi \sin 150t$

$$\left|\Delta\phi(t)\right|_{max} = \sqrt{(60\pi)^2 + (80\pi)^2} = 100\pi \text{ radian}$$

II) $\Delta\omega(t) = \dfrac{d\Delta\phi(t)}{dt} = -60\pi\,(150) + \sin(150t) + 80\pi \times 150 \cos(150t)$

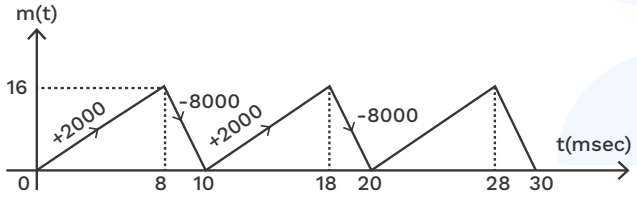
$$\left|\Delta\omega(t)\right|_{max} = \sqrt{\left(-60\pi(150)\right)^2 + (80\pi \times 150)^2}$$

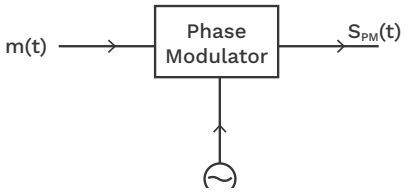
$$= 15\pi \frac{K\,rad}{sec}$$

III) $\beta = \dfrac{\left|\Delta\omega(t)\right|_{max}}{\omega_{max}} = \dfrac{\left|\Delta f(t)\right|_{max}}{f_{max}}$

$$\beta = \frac{15000\pi}{150} = 100\pi$$

Example:





1. Calculate K_p if maximum frequency deviation = 80 KHz
2. Maximum and minimum values of instantaneous frequency.

Solution:

1. $K_P : \dfrac{rad}{v}$

P.M. $\theta_i(t) = \omega_c t + K_P m(t)$

$$\omega_i(t) = \omega_c + K_P \frac{dm(t)}{dt}$$

$$\rightarrow f_i(t) = f_c + \frac{K_p}{2\pi}\frac{dm(t)}{dt}$$

$$\rightarrow f_i(t) = f_c + \Delta f(t)$$

$$\Delta f(t) = \frac{K_p}{2\pi}\frac{dm(t)}{dt}\ Hz$$

$$\Delta f(t) = \frac{K_p}{2\pi}\frac{dm(t)}{dt}$$

$$\left|\Delta f(t)\right|_{max} = 80KHz = \frac{K_p}{2\pi}\left|\frac{dm(t)}{dt}\right|_{max}$$

$$\frac{dm(t)}{dt} = \{2000, -8000\}$$

$$\left|\frac{dm(t)}{dt}\right| = \{2000, 8000\}$$

$$\left|\frac{dm(t)}{dt}\right|_{max} = 8000$$

$$80KHz = \frac{k_p}{2\pi} \times 8 \times 10^3$$

$$k_p = 20\pi\ \frac{rad}{v}$$

2. $\rightarrow \{f_i(t)\}_{max} = f_c + (\Delta f(t))_{max}$

$$= 1000KHz + \frac{K_p}{2\pi}\left\{\frac{dm(t)}{dt}\right\}_{max}$$

$$= 1000K + \frac{20\pi}{2\pi} \times (2000)$$

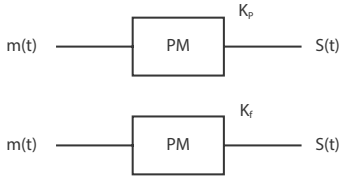
$$= 1000K + 20K = 1020K$$

$$\{f_i(t)\}_{min} = f_c + (\Delta f(t))_{min} = f_c + \frac{K_P}{2\pi}\left\{\frac{dm(t)}{dt}\right\}_{min}$$

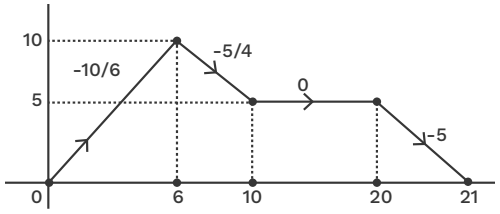
$$= f_c + \frac{20\pi}{2\pi}(-8000)$$

$$\{f_i(t)\}_{min} = 1000k - 80k = 920k$$

Example:



Calculate $\frac{K_p}{K_f}$ in $\frac{\text{rad}}{\text{Hz}}$ if maximum frequency deviation in both the cases is same.



Solution:

$$\frac{K_p}{K_f} : \frac{rad}{Hz} = \frac{rad\ /\ V}{Hz\ /\ v}$$

PM

$$K_p : \frac{rad}{v} : \quad \theta_i(t) = \omega_c t + K_p m(t)$$

$$\omega_i(t) = \omega_c + K_p \frac{dm(t)}{dt}$$

$$f_i(t) = f_c + \frac{K_p}{2\pi}\frac{dm(t)}{dt}$$

$$\Delta f(t)\big|_{max} = \frac{K_p}{2\pi}\left|\frac{dm(t)}{dt}\right|_{max} = \frac{5K_p}{2\pi} - i$$

FM

$$K_f : \frac{Hz}{Volt} \qquad f_i(t) = f + K_f m(t)$$

$$|\Delta f(t)|_{max} = K_f\ |m(t)|_{max} = 10K_f$$

$$\Delta f_{PM} = \Delta F_{FM}$$

$$\frac{5K_P}{2\pi} = 10K_f \rightarrow \frac{K_P}{K_f} = 4\pi\frac{rad}{Hz}$$

Test Your Understanding - 1

1. An angle-modulated signal is given by

$$s(t) = \cos 2\pi \begin{pmatrix} 2\times 10^6 t + 30\sin 150t \\ +40\cos 150t \end{pmatrix}$$

The maximum frequency and phase deviations of s(t) are

(A) $10.5\,kHz, 140\pi\,rad$

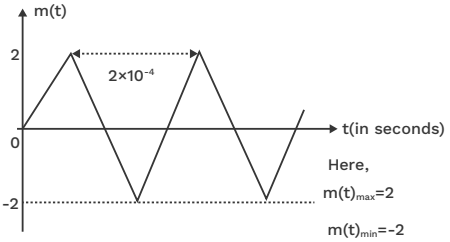
(B) $6\,kHz, 80\pi\,rad$

(C) $10.5\,kHz, 100\pi\,rad$

(D) $7.5\,kHz, 100\pi\,rad$

Solution: (D)

2. For the message signal shown below, calculate the maximum and minimum values of instantaneous frequency if the message signal is (A) phase modulated (B) frequency modulated.



The constant k_f *and* k_p are 2π × 105 rad/sec/V and $10\pi\,rad\,/\,V$ respectively, and the carrier frequency f_c is 100 MHz.

Solution:

1. $\left[f_i(t)\right]_{min} = 99.8\,MHz$

2. $\left[f_i(t)\right]_{min} = 99.8\,MHz$

Types of angle modulated signal:

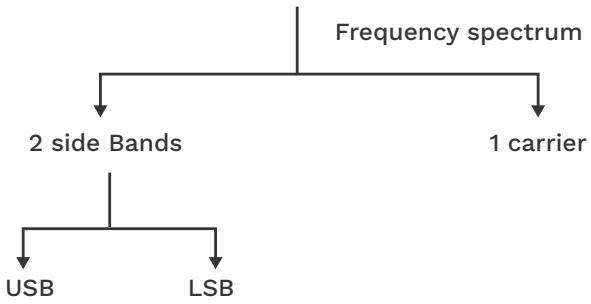
Types of FM:

1. Narrowband FM
2. Wideband FM

Types of PM:

1. Narrowband PM
2. Wideband PM

Nanowband Angle modulated signal:



Spectral analysis of angle modulated signal:

Case 1: m(t) is single tone sinusoidal signal.

$$\begin{cases} m(t) = A_m \cos 2\pi f_m t \\ c(t) = A_c \cos 2\pi f_c t \end{cases}$$

FM

$$S_{FM}(t) = A_c \cos\left(2\pi f_c t + \beta \sin 2\pi f_m t\right)$$

$$\updownarrow F.T.$$

$$S_{FM}(f) = ?$$

$$S_{FM}(t) = A_c \cos\left(2\pi f_c t + \beta \sin 2\pi f_m t\right)$$

$\theta \to small$

$\cos\theta \approx 1$

$\sin\theta \approx \theta$

Method 1:

$$S_{FM}(t) = A_c \cos\left(2\pi f_c t\right)\cos\left(\overbrace{\beta \sin 2\pi f_m t}^{Very\ small}\right)$$

$$-A_c \sin\left(2\pi f_c t\right)\sin\left(\overbrace{\beta \sin 2\pi f_m t}^{Very\ small}\right)$$

$\beta \to very\ small$:

$$S_{FM}(t) = A_c \cos 2\pi f_c t - A_c \sin\left(2\pi f_c t\right)\left(\beta \sin 2\pi f_m t\right)$$

$$S_{FM}(t) = A_c \cos 2\pi f_c t - A_c \beta \sin 2\pi f_c t \sin 2\pi f_m t$$

$$S_{FM}(t) = \underbrace{A_c \cos 2\pi f_c t}_{carrier} + \underbrace{\frac{A_c}{2}\beta \cos\left(2\pi\left(f_c + f_m\right)t\right)}_{USB}$$

$$\underbrace{-\frac{A_c \beta}{2}\cos\left(2\pi\left(f_c - f_m\right)t\right)}_{LSB}$$

$$S_{FM}(t) = S_{NBFM}(t) \to when\ \beta << 0.6 \to \beta <<<< 1$$

NBFM:

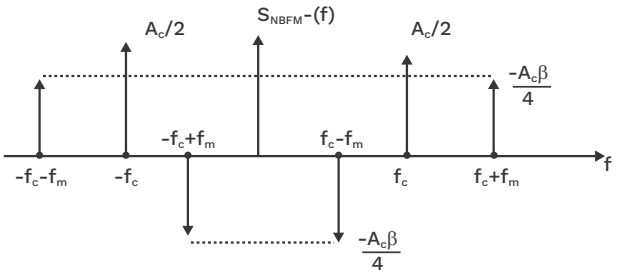
$$S_{FM}(t) = A_c \cos\left(\omega_c t + \beta \sin 2\pi f_m t\right)$$

$\beta <<< 1$

$$S_{FM}(t) = A_c \cos\omega_c t + \frac{A_c \beta}{2}\cos\left(2\pi\left(f_c + f_m\right)t\right)$$

$$-\frac{A_c \beta}{2}\cos\left(2\pi\left(f_c - f_m\right)t\right)$$

Spectrum:



Frequency related parameters:

1. $m(t) \to f_m$

   $c(t) \to f_c$

   $S_{NBFM}(t) \to f_c$

   $f_c \pm f_m$

2. $BW = 2f_m$

Power related calculation:

$$P_{NBFM} = P_c\left(1 + \frac{\beta^2}{2}\right)$$

$\beta \to very\ small$

$\beta^2 <<<<< 1$ $\qquad$ $P_{NBFM} \approx P_c \to \frac{A_c^2}{2}$

Relation between DSB-FC and NBFM:

$$S_{AM}(t) = A_c \cos 2\pi f_c t + \frac{A_c}{2}\mu \cos\left(2\pi\left(f_c + f_m\right)t\right)$$

$$+\frac{A_c}{2}\mu \cos\left(2\pi\left(f_c - f_m\right)t\right)$$

$$S_{NBFM}(t) = A_c \cos 2\pi f_c t + \frac{A_c \beta}{2}\cos\left(2\pi\left(f_c + f_m\right)t\right)$$

$$-\frac{A_c}{2}\beta \cos\left(2\pi\left(f_c - f_m\right)t\right)$$

1.

Freq. Comp.	Strength AM	Strength NBFM
f_c	$\frac{A_c}{2}$	$\frac{A_c}{2}$
$f_c + f_m$	$\frac{\mu A_c}{4}$	$\frac{\beta A_c}{4}$
$f_c - f_m$	$\frac{\mu A_c}{4}$	$\frac{-\beta A_c}{4}$

2. $\mu = \beta$

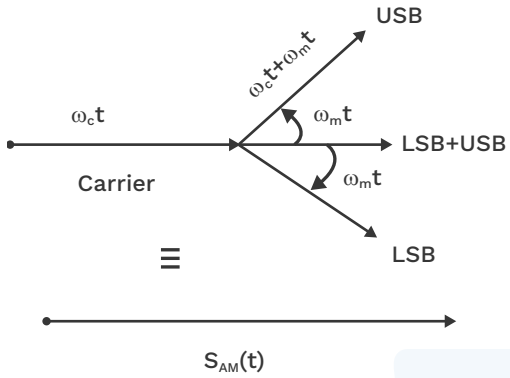
$$S_{NBFM}(t) + S_{AM}(t) = SSB - FC \rightarrow USB \rightarrow FC$$

$$S_{AM}(t) - S_{NBFM}(t) = SSB - FC \rightarrow LSB - SC$$

3. LSB in NBFM is 180° inverted w.r.t LSB in AM.

Phasor:

AM:



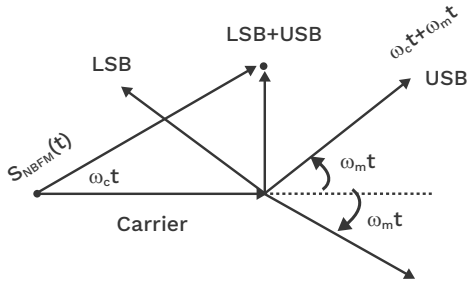
Length:

$A_c \rightarrow carrier$

$$\frac{A_c \mu}{2} \rightarrow USB$$

$$\frac{A_c \mu}{2} \rightarrow LSB$$

NBFM:

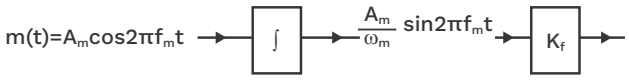


NBFM generation:

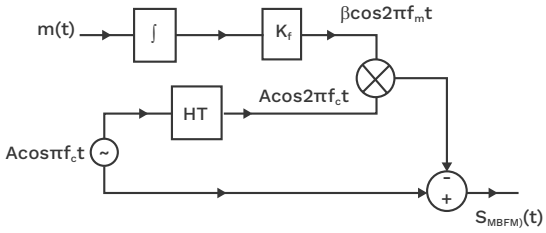
$$S_{NBFM}(t) = A_c \cos\left(2\pi f_c t + \beta \sin 2\pi f_m t\right)$$

≡

$$S_{NBFM}(t) = A_c \cos 2\pi f_c t - A_c \beta \sin 2\pi f_m + \sin 2\pi f_c t$$



$$\frac{k_f A_m}{\omega_m} \sin 2\pi f_m t = \beta \sin 2\pi f_m t$$



NBM modulator:

$$PM \rightarrow m(t) = A_m \cos\left(2\pi f_m t\right)$$

$$c(t) = A_c \cos\left(2\pi f_c t\right)$$

$$S_{PM}(t) = A_c \cos\left(\omega_c t + \beta \cos 2\pi f_m t\right)$$

$$\updownarrow F.T.$$

$$S_{PM}(f)$$

Method: $S_{PM}(t) = A_c \cos\left(\omega_c t + \beta \cos 2\pi f_m t\right)$

$$S_{PM}(t) = A_c \cos\omega_c t \underbrace{\cos\left(\beta \cos 2\pi f_m t\right)}_{\cos\theta \approx 1}$$

$$-A_c \sin\omega_c t \underbrace{\sin\left(\beta cos 2\pi f_m t\right)}_{\sin\theta \approx \theta}$$

$\beta <<< 1 \rightarrow small$

$$S_{PM}(t) = A_c \cos\omega_c t - A_c \beta \sin\omega_c t \cos\omega_m t$$

$$2\sin A\cos\beta = \sin\left(A+B\right) + \sin\left(A-B\right)$$

$$S_{PM}(t) = A_c \cos\omega_c t - \frac{A_c \beta}{2}\sin\left(\omega_c + \omega_m\right)t$$

$$-\frac{A_c \beta}{2}\sin\left(\omega_c - \omega_m\right)t$$

$$S_{PM}(t) = A_c \cos\omega_c t + \frac{A_c \beta}{2}\cos\left(\omega_c t + \omega_m t + 90°\right)$$

$$+\frac{A_c \beta}{2}\cos\left(\omega_c t - \omega_m t + 90°\right)$$

Frequency related parameters:

$m(t) \rightarrow f_m$

$c(t) \rightarrow f_c$

$s(t) \rightarrow f_c$

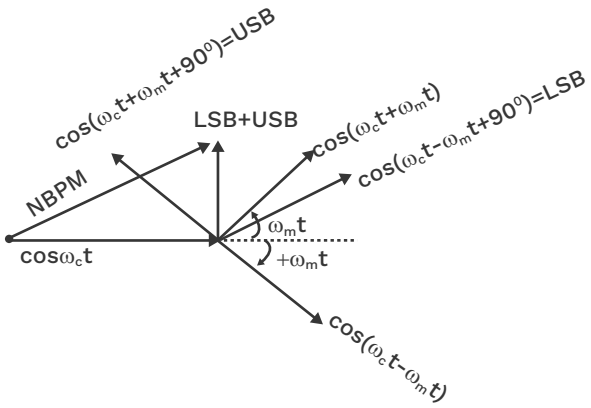
$f_c \pm f_m$

$BW = 2f_m$

Power related:

$$P_{NBPM} = P_c\left(1+\frac{\beta^2}{2}\right)$$

Phasor:



Conclusion

If m(t): Single tone sinusoidal.

Method 1:

S.1: Expand

S.2: $\beta << 1$

S.3: Narrow band angle modulated.

Note: If β becomes very large, then method 1 becomes in valid.

Method 2: For spectral analysis of angle modulated signals.

Case 1: $\rightarrow$ m(t) = Single tone sinusoidal signal

$$c(t) = A_c\cos 2\pi f_c t$$

$$m(t) = A_m\cos\omega_m t$$

$$S_{FM}(t) = A_c\cos\left(2\pi f_c t + \beta\sin 2\pi f_m t\right)$$

$$S_{PM}(t) = A_c\cos\left(2\pi f_c t + \beta\sin 2\pi f_m t\right)$$

$$m(t) = A_m\sin 2\pi f_m t$$

$$S_{FM}(t) = A_c\cos\left(2\pi f_c t - \beta\cos 2\pi f_m t\right)$$

$$S_{PM}(t) = A_c\cos\left(2\pi f_c t + \beta\sin 2\pi f_m t\right)$$

$$Sangle(t) = A_c\cos\left(2\pi f_c t + \beta\sin 2\pi f_m t\right)$$

$$\updownarrow F.T.$$

$$Sangle(f) = ?$$

$$Sangle(t) = A_c\cos\left(2\pi f_c t + \beta\sin 2\pi f_m t\right)$$

$$s(t) = \text{Re}\left\{\underbrace{A_c e^{j2\pi f_c t}}\ \underbrace{e^{j\beta\sin 2\pi f_m t}}\right\}$$

$$x(t) = e^{j\beta\sin 2\pi f_m t} \rightarrow periodic : T_m = \frac{1}{f_m}$$

$$\omega_m T_m = 2\pi$$

$$T_0 = T_m$$

$$\omega_0 = \omega_m$$

$$x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_m t}$$

$$C_n = \frac{1}{T_0}\int_{\frac{-T_0}{2}}^{\frac{T_0}{2}} x(t)e^{-jn\omega_m t} = \frac{1}{T_0}\int_{\frac{-1}{2f_m}}^{\frac{1}{2f_m}} x(t)e^{-jn\omega_m t}dt$$

$$C_n = \frac{1}{T_0}\int_{\frac{-1}{2f_m}}^{\frac{1}{2f_m}} e^{j\beta\sin 2\pi f_m t}\ e^{-jn\omega_m t}\ dt$$

$$C_n = \frac{1}{T_0}\int_{\frac{-1}{2f_m}}^{\frac{1}{2f_m}} e^{-j\left[-\beta\sin\omega_m t+\omega_m tn\right]}\ dt$$

$$C_n = \frac{1}{T_m\omega_m}\int_{-\pi}^{\pi} e^{-j\left[np-\beta\sin p\right]}\ dp$$

$$\omega_m t = p$$

$$\omega_m dt = dp$$

$$dt = \frac{dp}{\omega_m}$$

$$C_n = \frac{1}{2\pi}\int_{-\pi}^{\pi} e^{-j(np-\beta\sin p)}dp = J_n(\beta)$$

$\rightarrow$ Bessel's F^{n} of nth order

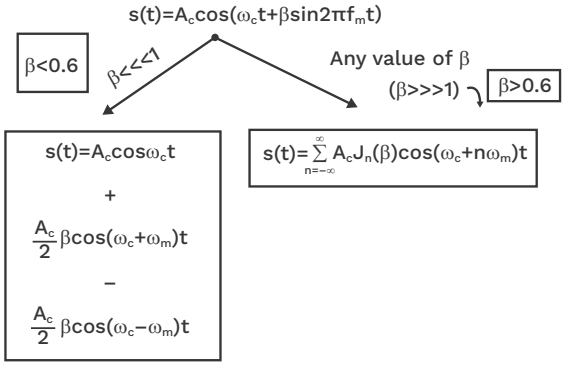
$$e^{j\beta\sin 2\pi f_m t} = x(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_m t} = \sum_{n=-\infty}^{\infty} J_n(\beta)e^{jn\omega_m t}$$

$$s(t) = \text{R}_e\left\{A_c e^{j\omega_c t}x(t)\right\}$$

$$= \text{R}_e\left\{A_c e^{j\omega_c t}\sum_{n=-\infty}^{\infty} J_n(\beta)e^{jn\omega_m t}\right\}$$

$$s(t) = R_e\left\{\sum_{n=-\infty}^{\infty} A_c J_n(\beta) e^{j(\omega_c + n\omega_m)t}\right\}$$

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos\left(2\pi(f_c + nf_m)t\right)$$



Spectral analysis of $s(t) = A_c \cos(2\pi f_c t + \beta \sin 2\pi f_m t)$

$$s(t) = \sum_{n=-\infty}^{\infty} A_c J_n(\beta)\cos\left(2\pi(f_c + nf_m)t\right)$$

Properties of Bessel's function:

1. $J_{-n}(\beta) = (-1)^n J_n(\beta)$

   $J_{-n}^2(\beta) = J_n^2(\beta)$

   $n: Even \qquad J_{-n}(\beta) = J_n(\beta)$

   $n: Odd \; J_{-n}(\beta) = -J_n(\beta)$

2. $\sum_{n=-\infty}^{\infty} J_n^2(\beta) = 1$

   $= J_0^2(\beta) + 2\left\{J_1^2(\beta) + J_3^2(\beta) + \ldots\right\}$

   $= J_0^2(\beta) + \left\{J_1^2(\beta) + J_{-1}^2(\beta)\right\}$

   $+ \left\{J_2^2(\beta) + J_{-2}^2(\beta)\right\} + \ldots$

3. $J.(\beta) = 0 : \beta = 2.4, 5.5, 8.6, 11.8$

4. $as\ n \uparrow : J_n(\beta) \downarrow$

   $s(t) = \sum_{n=-\infty}^{\infty} A_c J_n(\beta)\cos\left(2\pi(f_c + nf_m)t\right)$

   $A_c J_0(\beta)\cos 2\pi f_c t \rightarrow carrier\ component$

I^st order LSB

$A_c J_{-1}(\beta)\cos\left(2\pi(f_c - f_m)t\right)$ +

IInd order LSB

$A_c J_{-2}(\beta)\cos\left(2\pi(f_c - 2f_m)t\right)$ +

IIIrd order LSB

$A_c J_{-3}(\beta)\cos\left(2\pi(f_c - 3f_m)t\right)$

$\vdots$

tI^st order USB

$A_c J_1(\beta)\cos\left(2\pi(f_c + f_m)t\right)$

IInd order USB

$A_c J_2(\beta)\cos\left(2\pi(f_c + 2f_m)t\right)$

IIIrd order USB

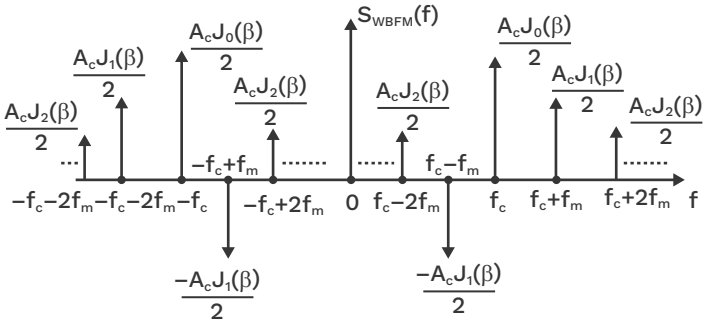
$A_c J_3(\beta)\cos\left(2\pi(f_c + 3f_m)t\right)$

$\vdots$

$If\ \beta <<< 1 \rightarrow s(t): 1\ carrier + 2SB \rightarrow NB\ Angle\ Mod^n$

$If\ \beta >>> 1 \rightarrow s(t): 1\ carrier + Infinite\ SB \rightarrow wideband\ Angle\ Mod^n$

Spectrum of wideband angle modulated signal:



Frequency related analysis:

$m(t) \rightarrow f_m$

$c(t) \rightarrow f_c$

$s(t) \rightarrow f_c$

$f_c \pm f_m \rightarrow If\ \ J_1(\beta) \neq 0$

$f_c \pm 2f_m \rightarrow If\ \ J_2(\beta) \neq 0$

$f_c \pm 3f_m \rightarrow If\ \ J_3(\beta) \neq 0$

Ideal BW of WBFM = ∞

Practical BW: Consider those sidebands which contain 98% of the power of time domain angle modulated signal.

$\Downarrow$

Carson's rule

Side Bands Considered Up to	Bandwidth	Only When
I order	$2f_m = 1 \times 2f_m$	$J_1(\beta) \neq 0$
II order	$4f_m = 2 \times 2f_m$	$J_2(\beta) \neq 0$
III order	$6f_m = 3 \times 2f_m$	$J_3(\beta) \neq 0$
⋮	⋮	
$(\beta+1)^{th}$ order	$(\beta+1)2f_m$ only when $J_{\beta+1}(\beta) \neq 0$	$J_{\beta+1}(\beta) \neq 0$

Bandwidth of angle modulated signal:

$BW = (\beta + 1)2f_m$

$\beta \rightarrow \beta_{PM} \rightarrow For\ \ PM$

$\beta \rightarrow \beta_{FM} \rightarrow For\ \ FM$

Spacing between adjacent spectral components = $\mathbf{f_m}$

Strength of impulses depends on $J_n(\beta)$

Frequency Component	Strength
f_c	$\frac{A_c J_0(\beta)}{2}$
$f_c + f_m$	$\frac{A_c J_1(\beta)}{2}$
$f_c - f_m$	$\frac{A_c J_{-1}}{2}(\beta)$
$f_c + 2f_m$	$\frac{A_c J_2}{2}(\beta)$
$f_c - 2f_m$	$\frac{A_c J_{-2}}{2}(\beta)$
⋮	⋮

Power related calculations:

m(t): Single tone sinusoidal $\rightarrow \frac{A_m^2}{2} = P_m$

$c(t): \frac{A_c^2}{2} = P_c$

Wideband angle modulated signal: $\overline{s^2(t)} = \frac{A_c^2}{2}$

$s(t) = A_c \cos(\omega_c t + \beta \sin 2\pi f_m t) \qquad \beta >> 1$

$$s(t) = \sum_{n=-\infty}^{\infty} A_c J_n(\beta) \cos(2\pi(f_c + nf_m)t)$$

$$s(t) = \underbrace{A_c J_0(\beta)\cos 2\pi f_c t}_{carrier} + A_c J_1(\beta)\cos(2\pi(f_c + f_m)t)$$
$$+ A_c J_{-1}(\beta)\cos(2\pi(f_c - f_m)t)$$
$$+ A_c J_2(\beta)\cos(2\pi(f_c + 2f_m)t)$$
$$+ A_c J_{-2}(\beta)\cos(2\pi(f_c - 2f_m)t)$$
$$+ \vdots$$
$$Sidebands$$

$$\overline{S^2(t)} = \frac{A_c^2}{2} J_0^2(\beta) + \frac{A_c^2}{2} J_1^2(\beta)$$
$$+ \frac{A_c^2}{2} J_{-1}^2(\beta) + \frac{A_c^2}{2} J_2^2(\beta) + \frac{A_c^2}{2} J_{-2}^2(\beta) + \ldots$$

$$\overline{S^2(t)} = \frac{A_c^2}{2} J_0^2(\beta) + A_c^2 J_1^2(\beta)$$
$$+ A_c^2 J_2^2(\beta) + A_c^2 J_3^2(\beta) + \ldots$$

$$\overline{s^2(t)} = \frac{A_c^2}{2}\left[J_0^2(\beta) + 2\left\{\begin{matrix} J_1^2(\beta) \\ +J_2^2(\beta) + J_3^2(\beta) + \ldots\ldots \end{matrix}\right\}\right]$$

$$\overline{s^2(t)} = P_c\left[J_0^2(\beta) + 2\left\{\begin{matrix} J_1^2(\beta) + J_2^2(\beta) \\ +J_3^2(\beta) + \ldots\ldots \end{matrix}\right\}\right]$$

Wideband angle modulated signal:

1. Power of carrier before modulation
$$= \frac{A_c^2}{2} = P_c$$
2. Power of carrier after modulation or power of angle modulated signal:
$$P = P_c\left[J_0^2(\beta) + 2\left\{\begin{matrix} J_1^2(\beta) + J_2^2(\beta) \\ +J_3^2(\beta) + \ldots\ldots \end{matrix}\right\}\right]$$
3. Power of carrier component in modulated signal:
$P_c J_0^2(\beta)$

4.

Power of Modulated Signal Up to	Power
I order SB	$P_c\left[J_0^2(\beta) + 2J_1^2(\beta)\right]$
II order SB	$P_c\left[J_0^2(\beta) + 2J_1^2(\beta) + 2J_2^2(\beta)\right]$
III order SB	$P_c\left[J_0^2(\beta) + 2\sum_{n=1}^{3} J_n^2(\beta)\right]$
⋮	⋮

5. Power of sidebands:
$$P_{SB} = 2P_c\left[J_1^2(\beta) + J_2^2(\beta) + \ldots\ldots\right]$$

6. $$\eta = \frac{P_{SB}}{P_{Total}} = \frac{2\left[J_1^2(\beta) + J_2^2(\beta) + \ldots\right]}{J_0^2(\beta) + 2\left\{J_1^2(\beta) + J_2^2(\beta) + \ldots\right\}}$$
$\beta = 2.4, 5.5, 8.6, 11.8$
$\rightarrow J_0(\beta) = 0 \Rightarrow \eta = 100\%$

7. If infinite side bands are considered:
$$P_{WB} = P_c\left[J_0^2(\beta) + 2\left\{J_1^2(\beta) + J_2^2(\beta) + \ldots\right\}\right]$$
$$P_{WB} = P_c\left\{\sum_{n=-\infty}^{\infty} J_n^2(\beta)\right\}$$
$$P_{WB} = P_c$$

Case 2: m(t) = Single tone

$c(t) = A_c \cos \omega_c t$

$$Sangle(t) = A_c \cos(\omega_c t + \beta \cos 2\pi f_m t)$$
$$\updownarrow$$
$$Sangle(f)$$

$s(t) = A_c \cos(\omega_c t + \beta \cos 2\pi_m t)$

$s(t) = R_e\left\{A_c e^{j\omega_c t} e^{j\beta \cos 2\pi f_m t}\right\}$

$x(t) = e^{j\beta \cos 2\pi f_m t} \rightarrow periodic\ T_m = \frac{1}{f_m}$

$T_m \omega_m = 2\pi$

$$x(t)=\sum_{n=-\infty}^{\infty} C_n e^{jn2\pi f_m t}$$

$$C_n=\frac{1}{T_m}\int_{\frac{-T_m}{2}}^{\frac{T_m}{2}} x(t)e^{-jn2\pi f_m t}dt$$

$$C_n=|C_n|e^{j\angle C_n}$$

$$x(t)=\sum_{n=-\infty}^{\infty}|C_n|e^{j\angle C_n}e^{j2\pi nf_m t}$$

$$x(t)=\sum_{n=-\infty}^{\infty}|C_n|e^{j(2\pi nf_m t+\angle C_n)}$$

$$s(t)=\sum_{n=-\infty}^{\infty}A_c|C_n|\cos(2\pi f_c t+2\pi nf_m t\ \angle C_n)$$

$$s(t)=\sum_{n=-\infty}^{\infty}A_c|C_n|\cos(2\pi(f_c+nf_m)t+\angle C_n)$$

Frequency related parameters:

$m(t)\rightarrow f_m$

$c(t)\rightarrow f_c$

$s(t)\rightarrow f_c \rightarrow If\ \ C_0\neq 0$

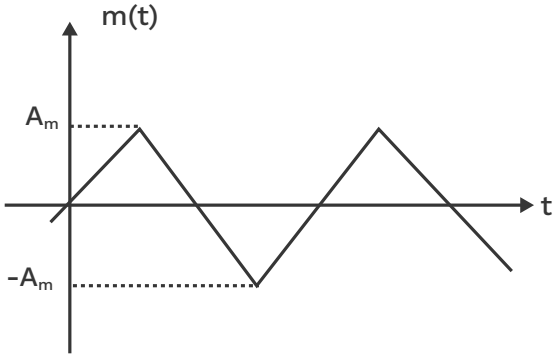
$f_c\pm f_m\rightarrow If\ \ C_1\neq 0$

$f_c\pm 2f_m\rightarrow If\ \ C_2\neq 0$

$\vdots$

$BW=(\beta+1)2f_m$: *only when* $C_{\beta+1}\neq 0$.

Case 3: m(t): Non-sinusoidal periodic signal



Infinite no. of discrete frequencies:

Let: $Sangle(t)=S_{PM}(t)=A_c\cos(\omega_c t+k_p m(t))$

S.1 $S_{PM}(t)=A_c\,\mathrm{Re}\{e^{j\omega_c t}e^{jk_p m(t)}\}$

S.2 $x(t)=e^{jk_p m(t)}$ $\qquad m(t)$: *periodic* $\begin{cases} I_m \\ f_m \end{cases}$

$$x(t)=\sum_{n=-\infty}^{\infty}|C_n|e^{j(2\pi f_m+\angle C_n)} \qquad T_m\omega_m=2\pi$$

$$S_{PM}(t)=\sum_{-\infty}^{\infty}A_c|C_n|\cos(2\pi f_c t+2\pi nf_m t+\angle C_n)$$

Freqⁿ: $m(t)\rightarrow$ *Infinite no. of discrete*

$c(t)\rightarrow f_c$

$s(t)\rightarrow f_c \rightarrow If\ \ C_0\neq 0$

$f_c\pm f_m\rightarrow If\ \ C_1\neq 0$

$f_c\pm 2f_m\rightarrow If\ \ C_2\neq 0$.

$\vdots$

$BW=(\beta+1)2f_m$

$$S_{FM}(t)=A_c\cos\left(\omega_c t+k_f\int m(t)dt\right) \qquad k_f:\frac{rad}{V-\sec}$$

$$a(t)=\int m(t)dt$$

$$S_{FM}(t)=A_c\cos(\omega_c t+k_f a(t))$$

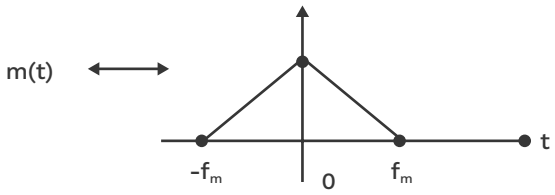
S.1 $S_{FM}(t)=\mathrm{Re}\left\{A_c\,e^{j\omega_c t}\underbrace{e^{jk_f a(t)}}_{x(t)}\right\}$

$$x(t)=e^{jk_f a(t)}=\sum|C_n|e^{j(2\pi nf_m t+\angle C_n)}$$

$$S_{FM}(t)=A_c\sum_{-\infty}^{\infty}|C_n|\cos(2\pi(f_c+nf_m)t+\angle C_n)$$

Freqⁿ analysis: Same as PM.

Case 4: m(t) is band limited signal:



$BW=(\beta+1)2f_{max}$

$f_{mav}=\max\ freq^n\ component\ of\ m(t)$

m(t)	Bandwidth
Sinusoidal single tone:	$(\beta+1)2f_m$
Non-sinusoidal periodic signal:	$(\beta+1)2f_m$ $\quad f_m$: *fundamental freq.*
Other cases for band limited m(t):	$(\beta+1)2f_{max}$ $\quad f_{max}$: max *freq^n component of* $m(t)$ = *Essential Band width*

PM $BW=(\beta_{PM}+1)2f_{max}$

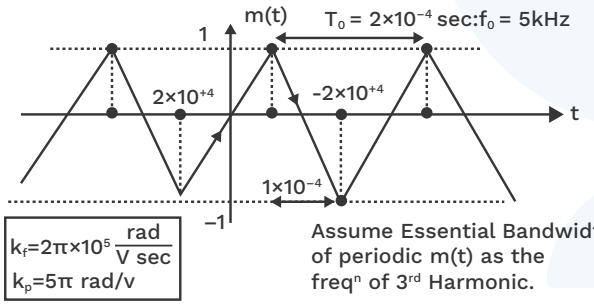
$\beta_{PM}=\dfrac{\Delta f}{f_{max}} \qquad \Delta f=\left|\Delta f(t)\right|_{max}$

$BW=2(\Delta f+f_{max})$

FM $BW=(\beta_{FM}+1)2f_{max}$

$BW=(\Delta f+f_{max})\times 2$

Example: Modulating signal is given as:



Assume Essential Bandwidth of periodic m(t) as the freq^n of 3rd Harmonic.

Calculate bandwidth of PM and FM = ?

Solution:

PM: $BW=(\beta+1)2f_{max} \qquad f_{max}=3f_0=15\,kHz$

$\beta=\dfrac{\Delta f}{f_{max}}=\dfrac{50}{15}\Rightarrow WBFM$

$k_p=5\pi\ \dfrac{rad}{v}\rightarrow\theta_i(t)=\omega_c t+k_p m(t)$

$f_i(t)=f_c+\dfrac{k_p}{2\pi}\dfrac{dm(t)}{dt}$

$\Delta f=\dfrac{k_p}{2\pi}\left|\dfrac{dm(t)}{dt}\right|_{max}=\dfrac{5\pi}{2\pi}\times2\times10^4=50\,kHz$

$BW=2(\Delta f+f_{max})=2(50+15)=130\,kHz$

FM: $BW=(\beta+1)2f_{max}=(\Delta f+f_{max})2$

$\beta=\dfrac{\Delta f}{f_{max}} \qquad f_{max}=15\,kHz$

$\Delta f=? \qquad k_f=2\pi\times10^5\ \dfrac{rad}{v\,sec}$

$\rightarrow\omega_i(t)=\omega_c+k_f m(t)$

$\Delta f=\dfrac{k_f}{2\pi}\left|m(t)\right|_{max}$

$\Delta f=100\,kHz$

$BW=(\Delta f+f_{max})\times2=230\,kHz$

Test Your Understanding - 2

1. Repeat previous question for PM = ?

Solution:

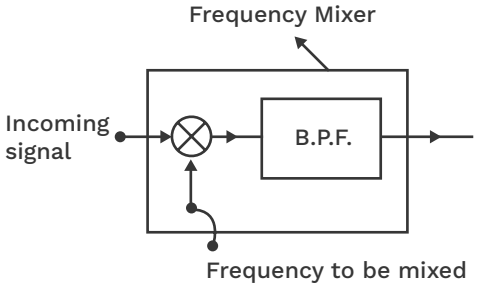
1. $\beta=4 \qquad B\omega=40KHz$

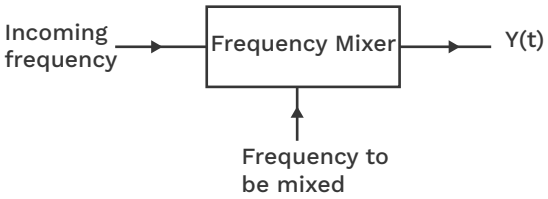
2. $A_{m_2}=2A_{m_1} \qquad \beta=\Delta\phi=K_pA_m=8$

$f_{m_2}=\dfrac{f_{m_1}}{4}$ $\boxed{B\omega=(\beta+1)2f_m=18KHz}$

Frequency Mixers and Frequency Multipliers

Frequency mixers:



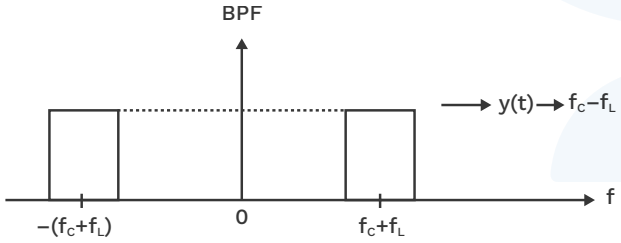


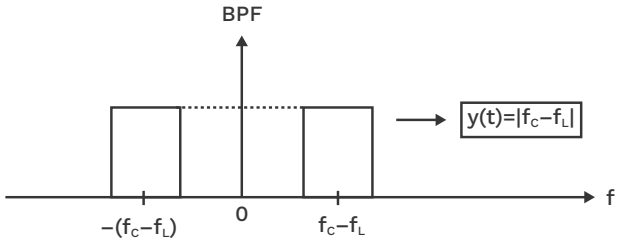
Let incoming signal: $\cos(2\pi f_c t)$

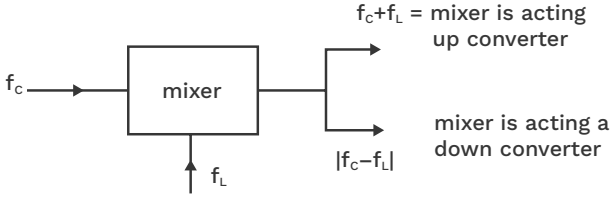
Frequency to be mixed: $\cos(2\pi f_L t)$

Output of multipliers: $\frac{1}{2}\cos(2\pi(f_c + f_L)t)$

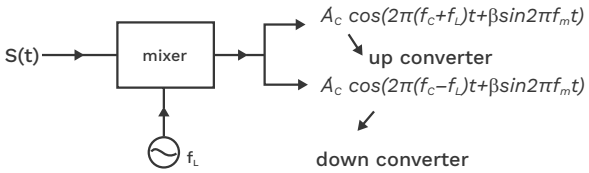
$\frac{1}{2}\cos(2\pi(f_c - f_L)t)$





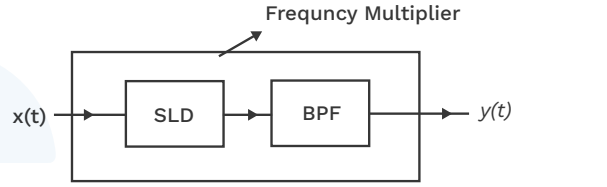


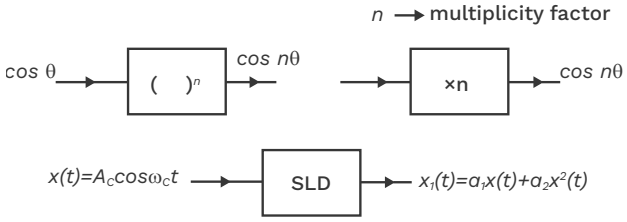
$$S(t) = A_c \cos\left(2\pi f_c t + \beta \sin 2\pi f_m t\right)$$



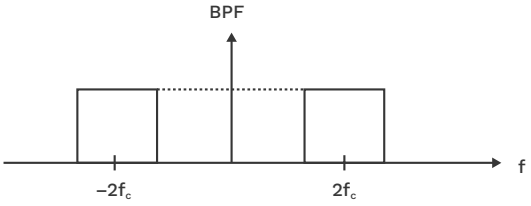
Mixer Input	Mixer Output
A_c	A_c'
f_c	$\lvert f_c - f_L \rvert$ or $f_c + f_L \rightarrow f_c$
β	β
f_m	f_m
Δf	Δf
BW	BW
Spectral spacing	f_m
Frequency Component	$f_c', f_c' \pm f_m, f_c' \pm 2f_m$

Frequency multiplier or angle multiplier:

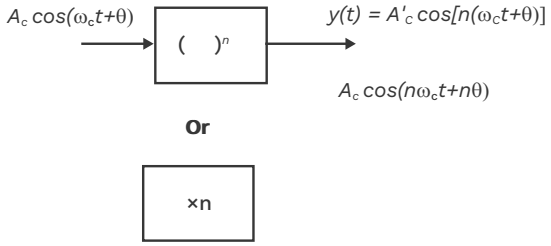




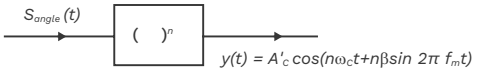
$$x_1(t) = a_1 A_c \cos\omega_c t + \frac{a_2 A_c^2}{2} + \frac{a_2 A_c^2}{2}\cos 2\omega_c t$$



$$y(t) = \frac{a_2 A_c^2}{2}\cos 2\omega_c t$$



$$S(t)A_c \cos\left(\omega_c t + \beta \sin\left(2\pi f_m t\right)\right)$$



Frequency Multiplier Input	Frequency Multiplier Output
A_c	A_c'
f_c	nf_c
f_m	f_m
β	$n\beta$
Δf	$n\Delta f$
BW	$(n\beta + 1)2f_m$
Spectral spacing	f_m
Frequency component	$(nf_c),(nf_c) \pm f_m,(nf_c) \pm 2f_m ..$

Example: 'In FM system a carrier of 100 MHz is modulated by a sinusoidal signal of 5 KHz. The bandwidth by Carson's approximation is 1 MHz'. If y(t) = (modulated waveform)^3, then by using Carson's approximation, the bandwidth of y(t) around 300 MHz and the spacing of spectral components are respectively.

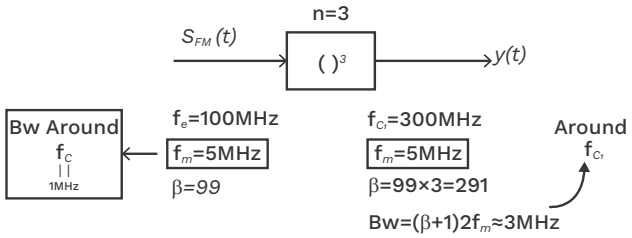
Solution:

$f_c = 100MHz$

$1000KHz = (\beta + 1)2f_m$

$f_m = 5KHz \qquad \beta = 99$

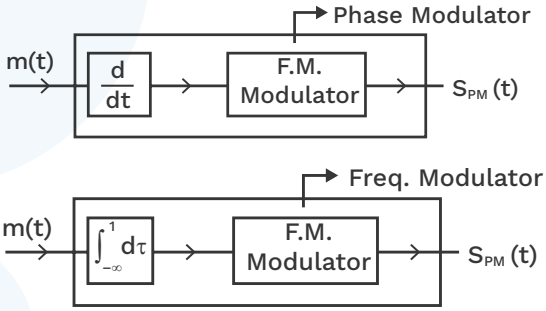
$BW = 1MHz$



Wideband angle modulation generation:

$$PM[m(t)] = FM\left[\frac{dm(t)}{dt}\right] \qquad If\ K_f = K_p = K$$

$$FM[m(t)] = PM\left[\int_{-\infty}^{t} m(\tau)d\tau\right]$$



Wideband FM generation:

1. Armstrong method (Indirect method)
2. Direct method

Indirect method or Armstrong method:

In this method, NBFM signal is converted to WBFM. So, WBFM is generated indirectly from NBFM by using frequency multiplication to increase deviation levels.

Block diagram:

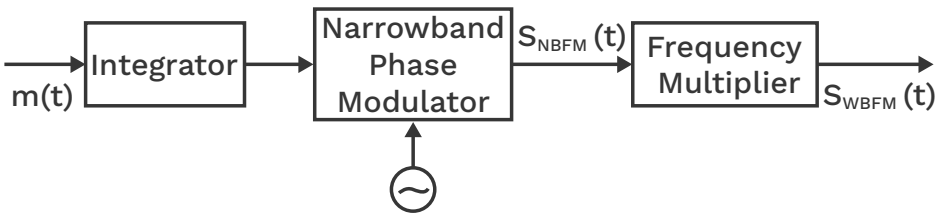
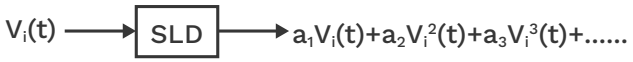


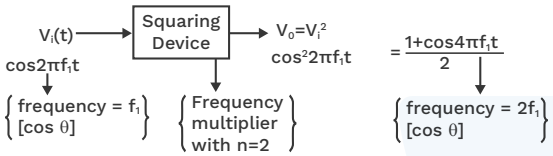
Figure-Block Diagram of Indirector Armstrong Method

- The message signal $m(t)$ is first integrated and then phase modulated.
- After generating $S_{WBFM}(t)$, a mixer can be used to change the frequency to 'f_c'.
- The modulation index β is small to reduce the distortion.
- A frequency multiplier is a square law device followed by a proper bandpass filter (BPF).

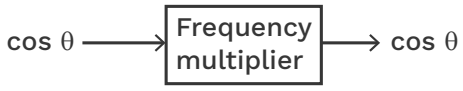
Square law device:



- The bandpass filter allows the component V_i^2 of the square law device to pass.



- So, frequency multiplier can be said as a square law device with multiplying constant (n) = 2



- The frequency multiplier is a non-linear device with no memory and has no energy storage elements.

The expression for single-tone FM is

$$S_{NBFM}(t) = A_c \cos\left\{2\pi f_c t + \left(\beta \sin 2\pi f_m t\right)\right\}; \beta < 1$$

And $\theta = 2\pi f_c t + \left(\beta \sin 2\pi f_m t\right)$

Which is passed through frequency multiplier (n).

$$S_{WBFM}(t) = A_c \cos\left\{2\pi n f_c t + n\beta \sin 2\pi f_m t\right\}$$

And $n\theta = 2\pi n f_c t + n\beta \sin 2\pi f_m t$

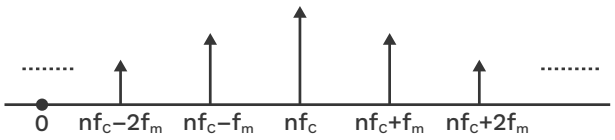
'n' should be such that $n\beta \geq 1$

Input of Frequency multiplier	*Output*
f_c	nf_c
β	$n\beta$
f_m	f_m
Δf	$n\Delta f$

Output of frequency multiplier (n)

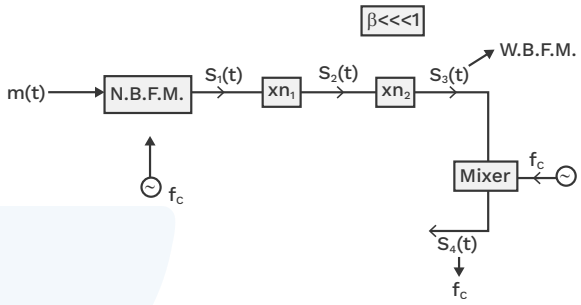
As, $\Delta f = \beta f_m$

So, when β increases by n, Δf also increases by n.



- Spacing between the frequency components remain unchanged.

Indirect (Armstrong) method:



$$S_1(t) = A_{c_1} \cos\left(2\pi\{f_c t + \beta \sin 2\pi f_m t\right)$$

$\beta <<< 1 NBFM$

$$S_2(t) = A_{c_2} \cos\left(2\pi n_1 f + n_1\beta \sin 2\pi f_m t\right)$$

$n_1\beta <<< 1 \quad NBFM$

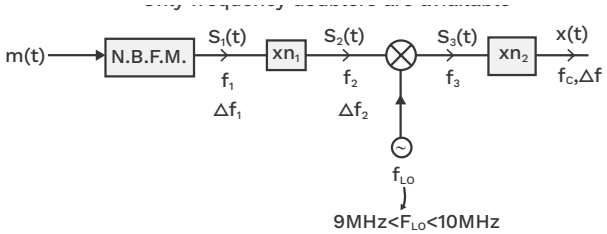
$$S_3(t) = A_{c_3} \cos\left(2\pi n_1 n_2 f_c + n_1 n_2 \beta \sin 2\pi f_m t\right)$$

$n_1 n_2 >> 1$ *WBFM*

$$S_4(t) = A_{c_4} \cos\left(2\pi\left(n_1 n_2 f_c \pm f_L\right) + n_1 n_2 \beta \sin 2\pi f_m t\right)$$

$$n_1 n_2 f_c \pm f_L = f_c \begin{array}{l}\nearrow Either = \\ \searrow |Mod \text{ of diff}|\end{array} OR$$

Example: Only frequency doublers are avaailable



Given:

$f_1 = 200 KHz$

$9\ Hz < \Delta f_1 < 10\ Hz$

Calculate: n_1, n_2 and $f_{Lo} = ?$

Solution:

$$\Delta f = n_1 n_2 \Delta f_1$$

$$\Delta f_1 = \frac{20,000}{n_1 n_2}$$

$$9Hz < \Delta f_1 < 10Hz$$

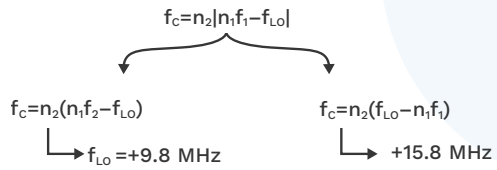
$$9Hz < \frac{20,000}{n_1 n_2} < 10Hz$$

$$n_1 n_2 = \underset{\text{Doublers were available}}{2^{11}} \rightarrow 2048 \quad \begin{matrix} n_1 = 2^6 = 64 \\ n_2 = 2^5 = 32 \end{matrix}$$

$$f_3 - \begin{vmatrix} \rightarrow n_1 f_1 + f_{Lo} \\ \rightarrow |n_1 f_1 - f_{Lo}| \end{vmatrix}$$

$f_c = 96MHz \quad f_1 = 200KHz$
$n_1 = 64$
$n_2 = 32$

$$f_c = n_2 \left(n_1 f_1 + f_{Lo}\right) \rightarrow f_{Lo} = -9.8MHz$$



Direct method:

- In direct method, the carrier frequency is directly varied in accordance with the message signal.
- This can be done by voltage-controlled oscillator (VCO).
- As discussed earlier, when the frequency of the carrier signal is modulated by the message, it is direct FM. While when phase of the carrier is modulated by message signal, it is direct PM indicating indirect FM.
- In direct FM, the frequency is varied or deviated directly by the amplitude of message signal. So, the instantaneous frequency deviation is directly proportional to the amplitude of message signal.



The tank circuit (L and C_a) shown below in simple direct FM generator determines the frequency of the LC oscillator.

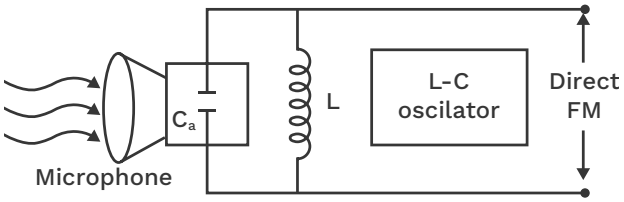


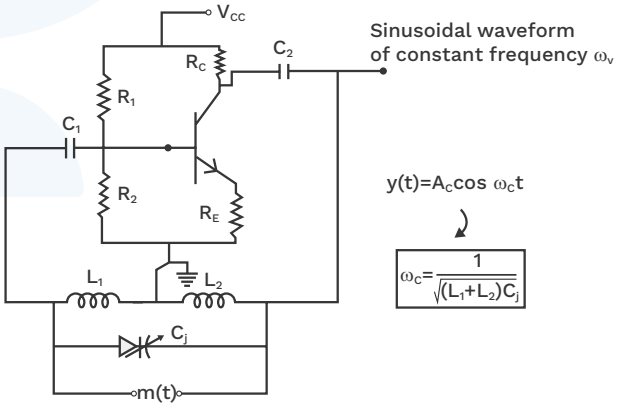
Figure: Simple circuit of direct FM Generation

The oscillator frequency varies directly according to external source such as sound, which is the message signal; therefore, the magnitude of the frequency change is proportional to amplitude of message signal (voltage).

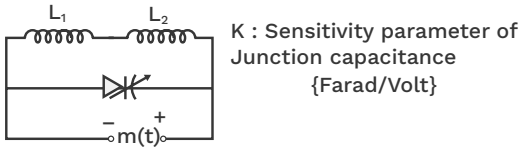
There are three ways of direct FM

1. Varactor diode method
2. FM eactance method
3. Linear IC direct FM method
   a) Voltage-controlled oscillator (VCO) is used to generate FM.
   b) VCO-modified version of Hartley oscillator:

Hartley oscillator:



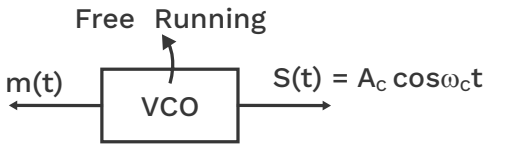
Hartley oscillator is converted into VCO if m(t) is applied. So, the name VCO.



$$m(t) \uparrow\uparrow \rightarrow R.B. \uparrow\uparrow \rightarrow \text{Depletion width} \uparrow\uparrow \rightarrow C_j = \frac{A \in}{d} \downarrow\downarrow$$

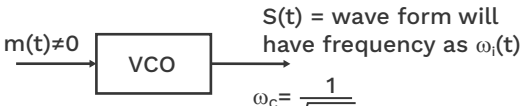
$$C_j = fm(t) \Rightarrow C_j(t) = C_o - km(t)$$

Important:



$\omega_C = \frac{1}{\sqrt{LC_1}}$ $L=L_1+L_2$

$C_1=C_0-k_m(t)=C_0$



$\omega_c = \frac{1}{\sqrt{LC_1 t}}$

$C_1(t)=C_j=C_0-km(t)$

$$\omega_i(t) = \frac{1}{\sqrt{LC_i(t)}}$$

$$\omega_i(t) = \frac{1}{\sqrt{L\left[C_0 - Km(t)\right]}}$$

$$\omega_i(t) = \frac{1}{\sqrt{LC_0\left(1-\frac{km(t)}{C_0}\right)}} \qquad \boxed{\frac{1}{\sqrt{LC_0}} = \omega_c}$$

$$\omega_i(t) = \omega_c\left\{1-\frac{Km(t)}{C_0}\right\}^{-1/2}$$

$$\boxed{\left|\frac{K_m(t)}{C_0}\right| <<< 1}$$

$$\omega_i(t) + 1 = \omega_c\left(1+\frac{K}{2}\frac{m(t)}{C_0}\right)$$

$$\omega_i(t) = \omega_c + \frac{k}{2}\frac{\omega_c}{C_0}m(t) = \omega_c + K_f m(t)$$

↓

Definition of FM:



$\omega_i(t) = \omega_i + K_f m(t)$

$\theta_i(t) = \omega_i + \int K_f m(t)$

$S(t) = A_i \cos(\omega_c t + \int K_f m(t))$

→ F.M.

$C_{junction}(t) = C_0 - km(t)$

C_0 : Residual capacitance

$\Delta C = Km(t)$ = Capacitance deviation

$$\omega_i(t) = \omega_c + \frac{k\omega_c}{2C_0}m(t)$$

$\omega_c \rightarrow$ *carier frequency*

$$\Delta\omega = \frac{K\omega_c}{2C_0}m(t) = \frac{\omega_c}{2}\left\{\frac{Km(t)}{C_0}\right\}$$

→ Frequency deviation

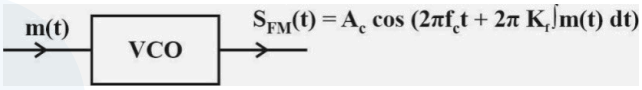
Remember:

$$\frac{\Delta\omega}{\omega_c} = \frac{Km(t)}{2C_0} = \frac{\Delta c}{2C_a}$$

$$\frac{\Delta\omega}{\omega_c} = \frac{\Delta C}{2C_0}$$

$$\left|\frac{Km(t)}{C_0}\right| <<< 1 \rightarrow \quad \frac{\Delta C}{C_0} <<< 1$$

$$\frac{\Delta\omega}{\omega_c} = \frac{\Delta C}{2C_0} <<< \frac{1}{2}$$



There are various types of detectors used to demodulate FM signal. They are

Frequency discriminators:

1. Simple slope detector
2. Balanced slope detector

Phase discriminators:

1. Foster–Seeley detector
2. Ratio detector

Phase locked loop (PLL)

Zero crossing detector:

Frequency and phase discriminators require transformers for their construction. So, for demodulation, generally PLL method is preferred.

Frequency discriminators:

Simple slope detector:

Circuit diagram:

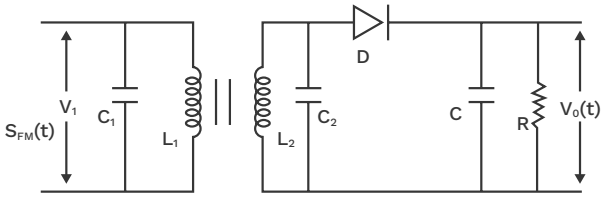
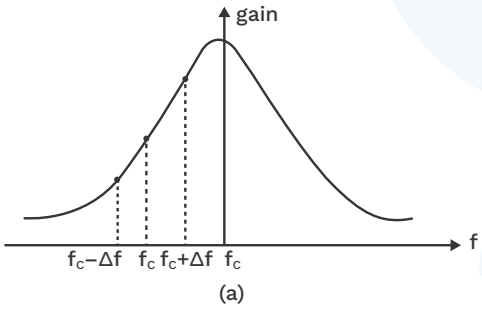


Figure: Circuit of Simple Slope Detector

- A device which gives a response proportional to the frequency deviation of the input signal is called a frequency discriminator (or slope detector).
- Here, the transformer acts as differentiator and diode (with RC) as envelope detector.
- The tank circuit (or transformer) is tuned at a resonant frequency Fr, which is more than maximum frequency of FM signal.
- The tuned circuit is non-linear.
- The characteristics are



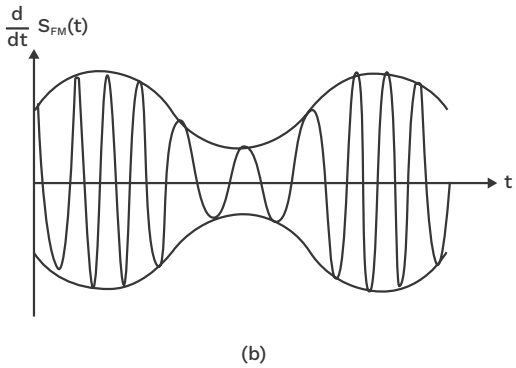


Figure: Characteristics of Simple Slope Detector (b) Waveform at output of Differentiation

- Linearity of this characteristic curve is the useful zone.
- The linear portion has a slope, which works as a differentiator.
- The transformer (differentiator) followed by a nonlinear device, i.e., diode (envelope detector) is the frequency discriminator.

Drawbacks:

- Useful range is limited.
- Due to non-linear characteristics, there is a variation in amplitude which is not reconstructed perfectly (slope error).
- Tuning of two tank circuits is difficult.

Balanced slope detector:

- The drawback of simple slope detector is limited linear region that is partially rectified in balanced slope detector.
- We will be using three-tuned circuits, with a centre-tapped transformer.
- Hence, the frequency discriminator has a pair of slope circuits followed by envelope detectors.
- The tuning range is increased; it is called a staggered tuned circuit.

Circuit diagram:

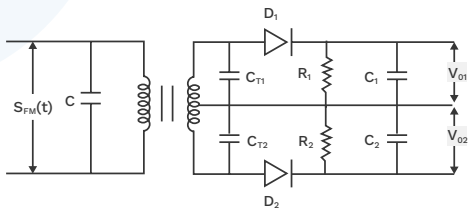


Figure: Circuit Diagram of Balanced Slope Detector

The primary transformer is tuned at f_c, the upper tank circuit is tuned at $f_c + \Delta f$ and the lower tank circuit is tuned at $f_c - \Delta f$, where Δf is the frequency deviation.

- D1 conducts for positive half-cycle of the input signal (positive envelope) and D2 conducts for the negative half-cycle of the input signal (negative envelope).
- Characteristics of the discriminator are

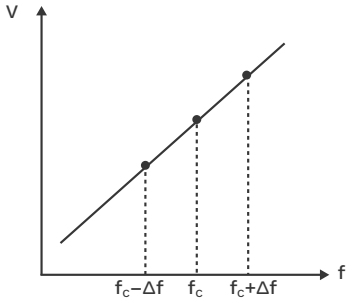


Figure: (a) Ideal output voltage Characteristics

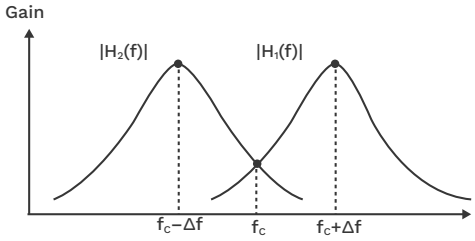


Figure: (b) Characteristics of the two Filters

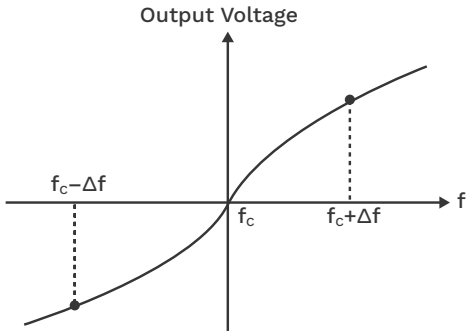


Figure: (c) S-shaped characteristics (Output)

Case 1:

When the circuit is tuned at f_c

$V_{01} = V_{02}$

So, $V_0 = V_{01} - V_{02} = 0$

The output voltage is zero.

Case 2:

When the circuit is tuned at $f_c + \Delta f$

Then $V_{01} > V_{02} = 0$

$\therefore V_0 = V_{01} - V_{02}$

The output voltage is positive.

Case 3:

When the circuit is tuned of $f_c - \Delta f$

Then $V_{01} < V_{02}$

$\therefore V_0 = V_{01} - V_{02}$

The output voltage is negative.

Drawbacks:

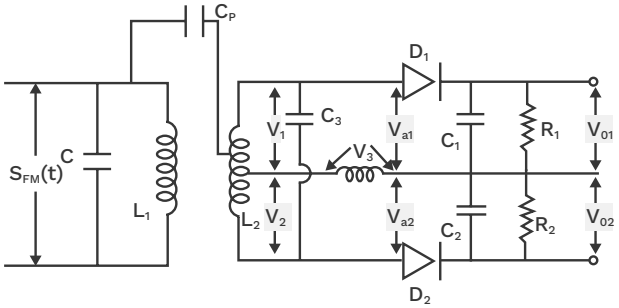
- Three tank circuit at different frequencies are difficult to tune.
- The linear characteristics are limited.
- It is sensitive to amplitude variations in the input signal (FM signal).

Phase Discriminators

In this method, we will focus on a very popular method of demodulation of FM signal, i.e., Foster–Seeley method of detection.

Foster–Seeley method:

- In phase discriminator method, the phase difference in the output of tuned circuit is used to calculate the amplitude variation.
- **Circuit diagram:**



- The difference in the circuits of the Foster–Seeley and balanced slope is the coupling capacitor (C_p) and the tuning circuits are tuned at same frequency (f_c).
- The voltages V_1 and V_2 are out of phase to each other.
- The output voltage is

$$V_0 = |V_{02}| - |V_{01}|$$

The voltages after inductor L_3 are

$$V_{a1} = V_3 - V_1 \quad V_{a2} = V_3 - V_2.$$

- The characteristics are

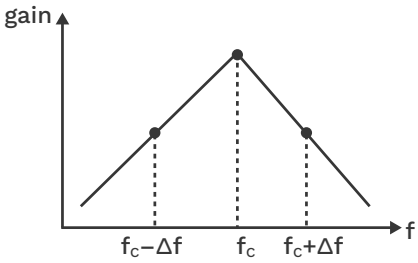


Figure: Characteristics of Foster-Seeley Method

Case 1: At resonance f_c

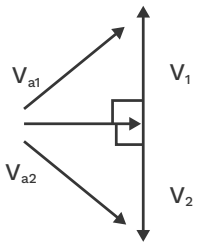


Figure (a): Phaser diagram at f_c

Now, as $|V_{a1}| = |V_{a2}|$

So, $V_0 = |V_{02}| - |V_{01}| = 0$

Case 2:

At frequency, $f_c + \Delta f$

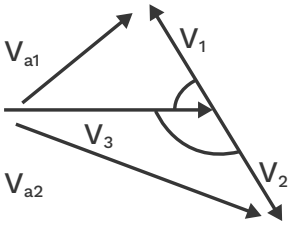


Figure: (b) Phaser diagram at $f_c + \Delta f$

Here, $|V_{a1}| < |V_{a2}|$

$\therefore V_0 = |V_{02}| - |V_{01}|$ is positive.

Case 3:

At frequency, $f_c - \Delta f$ m

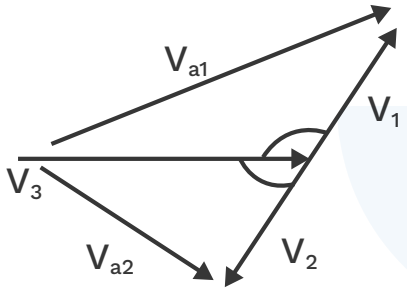


Figure: (c) Phaser diagram at f_c - Δf

Here, $|V_{a1}| > |V_{a2}|$

$\therefore V_0 = |V_{02}| - |V_{01}|$ is negative.

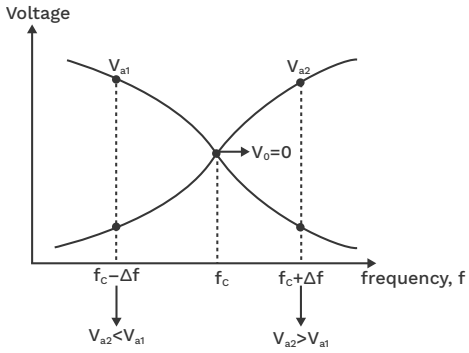


Figure: S-shaped Characteristics of Foster-Seeley Detector

Advantages:

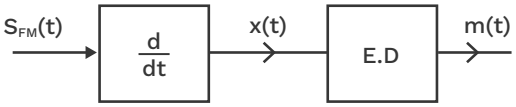
- Offers a much better linearity.
- Easy to construct.

Drawbacks:

- Not compatible with IC technology.
- Amplitude noise is present, so we need an amplitude limiter circuit before the discriminator.
- It is costly.

FM demodulation:

Theoretical method:



$$S_{FM}(t) = A_c \cos\left(2\pi f_c t + 2\pi K_f \int m(t)\,dt\right)$$

$$x(t) = \frac{dS_{FM}(t)}{dt} = -\left(2\pi f_c + 2\pi K_f m(t)\right)$$
$$\left(A_c \sin(2\pi f_c t + 2\pi K_f \int m(t)\,dt\right)$$

$$x(t) = -A_c\left(2\pi f_c + 2\pi K_f m(t)\right)$$
$$\sin\left(2\pi f_c t + 2\pi K_f \int m(t)dt\right]$$

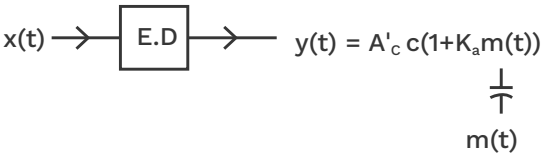
$$x(t) = -A_c\left(2\pi f_c\right)\left[1 + \frac{K_f}{f_c} m(t)\right]$$
$$\sin\left(2\pi f_c t + 2\pi K_f \int m(t)dt\right)$$

Let $-A_c 2\pi f_c = A_c' \qquad K_f / f_c = K_a$

$$x(t) = A_c'\left(1 + K_a m(t)\right)\sin\left(2\pi f_c t + 2\pi K_f \int m(t)dt\right)$$

Angle amplitude modulated signal

$$\mu = K_a \left|m(t)\right|_{max} = \frac{K_f}{f_c}\left|m(t)\right|_{max} = \frac{\Delta f}{f_c}$$

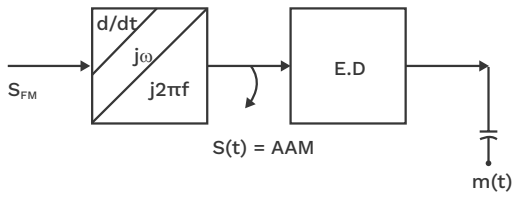


Condition:

$\mu \le 1$

$$\frac{k_f}{f_c}\left|m(t)\right|_{max} \le 1$$

Summary

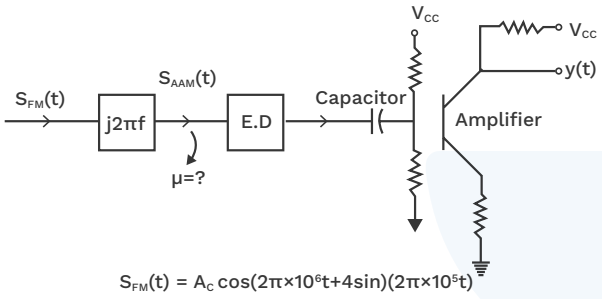


$$\mu = \frac{\Delta f}{f_c} \qquad K_a = \frac{K_f}{f_c}$$

$$\Delta f = K_f \mid m(t) \mid_{max}$$

$$\mu \leq 1$$

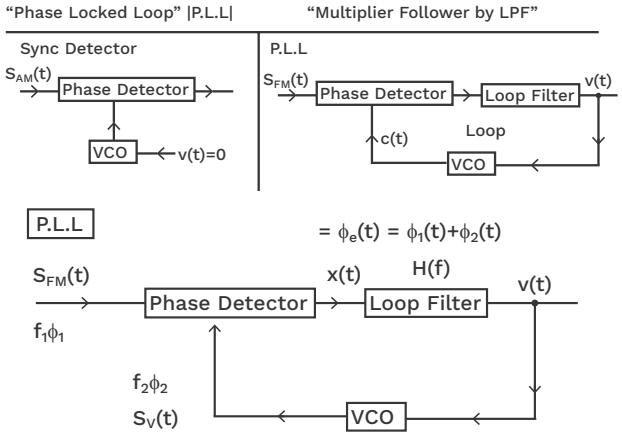
Example:



Will $y(t) \alpha m(t) = ? \rightarrow Yes$

$$\mu = \frac{\Delta f}{f_c} = \frac{\beta f_m}{f_c} = \frac{4 \times 10^5}{10^6} = 0.4$$

Practical FM demodulator:



Purpose → $v(t) \alpha\ m(t) \rightarrow$ Proof

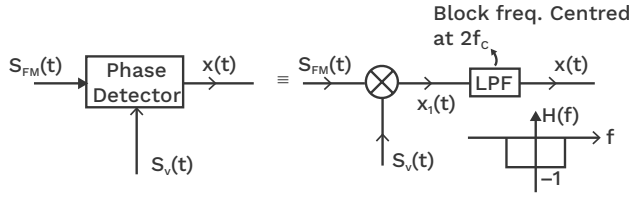
Blocks of PLL:

Input of PLL is frequency modulated signal

$$S_{FM} = A_c \cos\left(\omega_c t + \phi_1(t)\right)$$

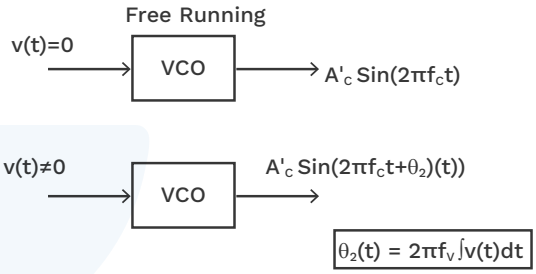
$$\phi_1(t) = 2\pi K_f \int m(t)\,dt$$

Phase comparator: Frequency multiplier followed by LPF.



Loop filter: It has to be designed such that $v(t) \alpha\ m(t)$.

VCO: Sinusoidal waveform generator whose frequency varies linearly with message signal m(t).



Mathematical analysis:

$$S_{FM}(t) = A_c \cos\left(\omega_c t + \phi_1(t)\right)$$

$$\phi_1(t) = 2\pi K_f \int m(t)\,dt$$

$$S_V(t) = A_c' \sin\left(\omega_c t + \phi_2(t)\right) \quad \uparrow \text{Frequency Lock}$$

$$\phi_2(t) = 2\pi K_V \int m(t)\,dt$$

Output of multiplier:

$$x_1(t) = \frac{A_c A_c'}{2} \sin\left(2\omega_c t + \phi_1(t) + \phi_2(t)\right)$$

$$+ \frac{A_c A_c'}{2} \sin\left(\phi_2(t) - \phi_1(t)\right)$$

Output of LPF:

$$x(t) = -\frac{A_c A_c'}{2} \sin\left(\phi_2(t) - \phi_1(t)\right)$$

$$x(t) = \frac{A_c A_c'}{2} \sin\left(\phi_1(t) - \phi_2(t)\right)$$

$\theta \rightarrow$ small
$\sin\theta \rightarrow \theta$
$\cos\theta \rightarrow 1$

$\phi_1(t) - \phi_2(t) = \phi_e(t) \rightarrow$ Phase Error

Phase lock: $\phi_e(t) \rightarrow 0 \;\; x(t) = \phi_1(t) - \phi_2(t)$

if $\frac{A_c A_c'}{2} = 1$

$x(t) = \phi_e(t) \rightarrow 0$

$x(t) = \phi_1(t) - \phi_2(t) = \phi_e(f)$

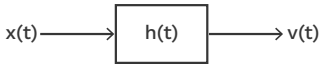
$x(t) = 2\pi K_f \int m(t)dt - 2\pi K_v \int v(t)dt$

$\frac{dx(t)}{dt} = 2\pi K_f m(t) - 2\pi K_v v(t)$

Fourier transform on both sides:

$j2\pi f \;\; X(f) = 2_\pi K_f M(f) - 2\pi K_v V(f)$

$if \;\; X(f) = K_f M(f) - K_v V(f)$ ...(i)



$v(t) = x(t) * h(t)$

$v(f) = x(f)H(f)$ ...(ii)

Put (ii) in (i)

If $X(f) = K_f M(f) - K_v X(f) H(f)$

$X(f) = \frac{K_f M(f)}{jf + K_v H(f)}$

$V(f) = \frac{H(f) K_f M(f)}{jf + K_v H(f)}$

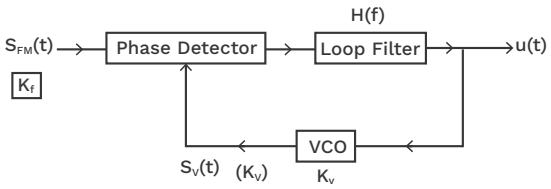
Design of loop filter:

$|H(f)| \rightarrow$ Very high

$V(f) = \frac{H(f) K_f M(f)}{jf\left[1 + \frac{K_v}{jf} H(f)\right]} = \frac{H(f) K_f M(f)}{jf \frac{K_v}{jf} \times H(f)}$

$V(f) = \frac{K_f}{K_v} M(f) \rightarrow v(t) = \frac{K_f}{K_v} m(t)$

Summary



1. $v(t) = \frac{K_f}{K_v} m(t)$
2. Lock mode → Frequency lock
3. Capture mode → Phase lock
4. Lock range ≥ Capture range

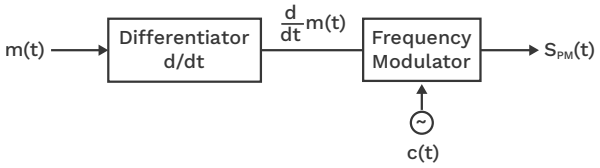
Similarity of PM with FM:

The expressions of PM and FM respectively

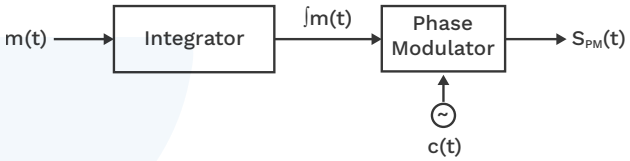
$S_{PM}(t) = A_c \cos\left[2\pi f_c t + K_p m(t)\right]$

$S_{FM}(t) = A_c \cos\left[2\pi f_c t + 2\pi K_f \int m(t)dt\right]$

Generation of PM from FM:



Generation of FM from PM:



Conversion can be understood by:

Frequency modulation:

$FM\left[m(t)\right] = PM\left[\int m(t)\right]$

Phase modulation:

$PM\left[m(t)\right] = FM\left[\frac{d}{dt} m(t)\right]$

Maximum Phase Deviation of FM Signal

We have earlier seen phase deviation of PM signal. Now, we will see for FM. From the general expression of FM signal given below, the phase deviation is

$S_{FM}(t) = A_c \cos\left(2\pi f_c t + 2\pi k_f \int m(t)dt\right)$

$(\Delta\phi)_{FM} = \left|\max\left(2\pi k_f \int m(t)dt\right)\right|$

It is similar as how to find phase deviation of PM from general expression of PM signal above.

$(\Delta\phi)_{FM} = \left|\max\left[k_p m(t)\right]\right|$

When $m(t) = A_m \cos 2\pi f_m t$

$$(\Delta\phi)_{FM} = \left|\max\left(2\pi k_f \times \frac{A_m \sin 2\pi f_m t}{2\pi f_m}\right)\right|$$

$$= \frac{k_f A_m}{f_m} = \beta_{FM}$$

$$\therefore (\phi)_{FM} = \beta_{FM}$$

And $(\phi)_{PM} = k_p A_m$

Maximum Frequency Deviation of PM Signal

As we have seen frequency deviation of FM signal, now we will see for PM. From the general expression of PM signal given below, the frequency deviation is

$$S_{PM}(t) = A_c \cos\left(2\pi f_c t + K_p m(t)\right)$$

Where $\theta_i = 2\pi f_c t + K_P m(t)$

$$S_{PM}(t) = A_c \cos\left(\theta_i(t)\right)$$

Instantaneous frequency is

$$f_i = \frac{1}{2\pi}\frac{d}{dt}\theta_i(t)$$

$$f_i = \frac{1}{2\pi}\left(2\pi f_c + k_p \frac{d}{dt} m(t)\right)$$

$$\left[f_i = f_c + \text{frequency deviation or } \Delta f\right]$$

$$f_i = f_c + \frac{k_p}{2\pi}\frac{d}{dt} m(t)$$

$$\therefore (\Delta f)_{PM} = \left|\max\left(\frac{k_p}{2\pi}\frac{d}{dt} m(t)\right)\right|$$

$$\&\,(\Delta f)_{FM} = \left|\max\left(k_f m(t)\right)\right|$$

When $m(t) = A_m \cos 2\pi f_m t$

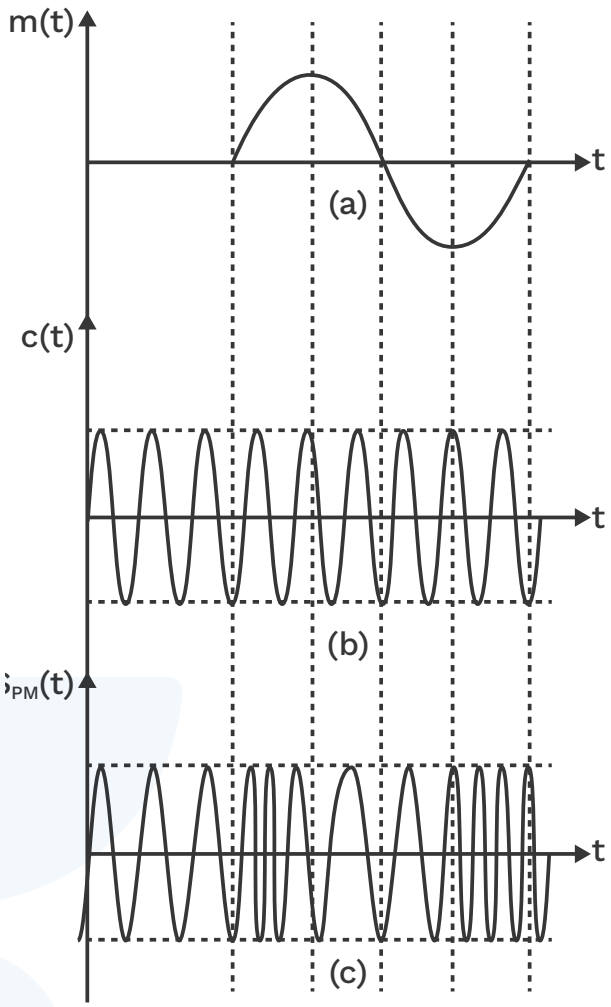
$$(\Delta f)_{PM} = \left|\max\left(\frac{k_p}{2\pi}.\left(-A_m 2\pi f_m \sin 2\pi f_m t\right)\right)\right|$$

Therefore,

$$(\Delta f)_{PM} = k_p A_m f_m = \beta_{PM}.f_m$$

$$(\Delta f)_{PM} = k_f A_m$$

Waveforms of Phase Modulation



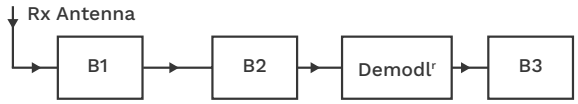
- FM and PM waveforms can be distinguished by observing the message signal, as both look similar except the time or phase relation.
- With FM, the maximum frequency deviation occurs during maximum positive and negative peaks of m(t).
- That means the frequency deviation depends on amplitude of message signal.
- For PM, the maximum frequency deviation occurs during zero crossings of m(t).
  That means the frequency deviation depends on the slope [derivative of m(t) or $\frac{d}{dt} m(t)$].

Superheterodyne Rx

Difference between Rx and demodulator:

1. Demodulator is part of receiver

2.



Analog receivers:

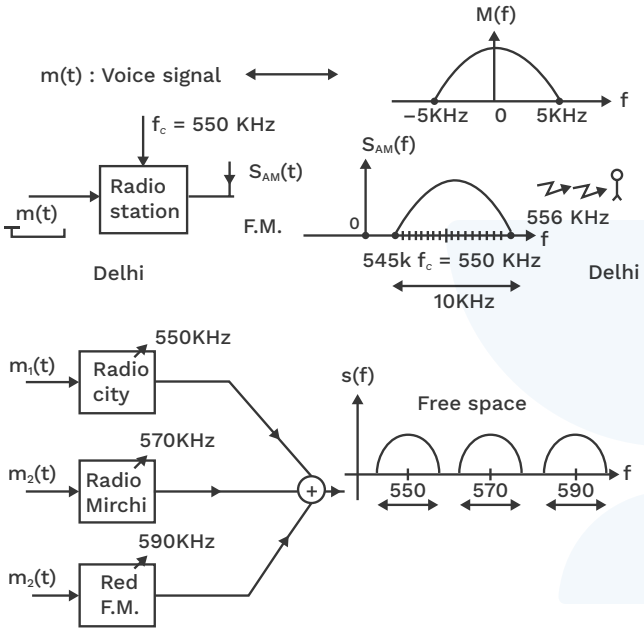
1. Receives analog modulated signals:

E.g., AM, FM, PM

A.M.	*F.M.*
$f_c \Rightarrow 535\,KHz - 1640\,KHz$	$f_c \Rightarrow 88\,MHZ - 108MHz$
$m(t) \Rightarrow 10\,KHz$	$m(t) \Rightarrow 200\,KHz$
$I.F. \rightarrow 455\,KHz$	$I.F. \Rightarrow 10.7\,MHz$

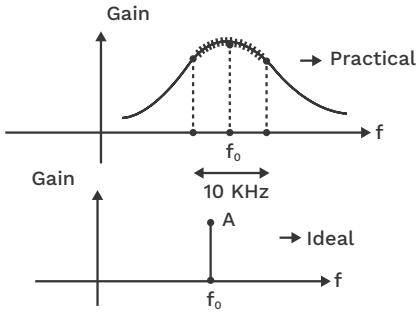
AM modulation scheme:

RJ:



Note: *'The complete AM spectrum is present at the input of superheterodyne receiver'.*

Tuned amplifier:



Receivers:

- The receiver in any type of broadcasting system like those based on amplitude or frequency modulation has the role of performing various other tasks apart from demodulating the input signal. Some of these tasks are:
  - Carrier frequency tuning: Its purpose is to select desired signal.
  - Filtering: It is required to separate the desired signal from other modulated signals which may be picked up along the way.
  - Amplification: It is intended to compensate for the loss of signal power incurred during the course of transmission.

Characteristic parameters of receivers:

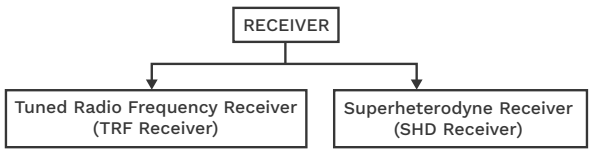
- **Sensitivity**: It is defined as the minimum signal that should be applied at the input of the receiver to get the standard output. Sensitivity of receivers is directly proportional to the gain of the amplifier, i.e., if gain of the amplifier is high, the sensitivity is also high.
- **Selectivity**: It is ability of the receiver to reject the unwanted components and select the desirable components.
- **Fidelity**: It is the ability of the receiver to produce all frequency components of transmitted message signal.

E.g., The transmitter is transmitting (30 Hz - 30 KHz) band but receiver is producing (35 Hz - 20 KHz). The receiver is not able to produce all the frequencies. Hence, fidelity of receiver is low.

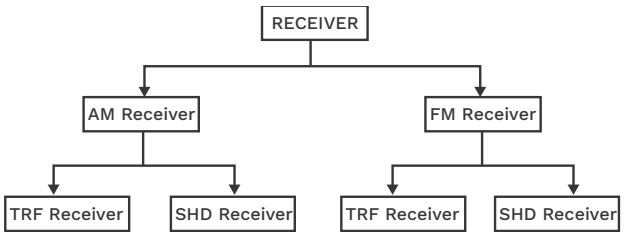
If the receiver produces (30 Hz - 30 kHz) band, fidelity of receiver is high.

Classification of receivers:

On the basis of construction:



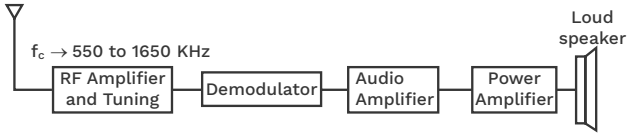
On the basis of modulation scheme:



AM receivers:

- An AM receiver is made to receive and detect an amplitude modulated signal.
- An amplitude modulated system varies the amplitude of the carrier with respect to amplitude changes of a modulating signal.

Tuned Radio Frequency Receiver (TRF Receiver)



Tuned radio frequency (TRF) receiver is a receiver where the tuning, i.e., selectivity is provided by the radio frequency stages.

RF amplifier and tuning:

This stage consists of one or more amplification and tuning stages. This stage is basically used to provide some gain and selectivity to the receiver.

Demodulator:

Demodulator is used to extract message signals from the amplitude modulated signals stage generally uses envelope detector to extract original message signal.

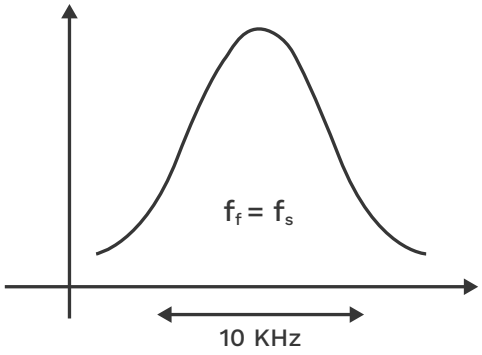
Audio amplifier:

This stage is used to provide audio amplification.

Power amplifier:

- Power amplifier is generally used to provide power to audio devices, e.g., speakers.
- It does not have any extra features and connections.
- We often find this in a home theatre receiver such as radio, input source switching and audio video processing.
- Bandwidth allotted for each channel in AM signal is 10 KHz according to ITU (International Telecommunication Union).

$$B.W = \frac{f_r}{Q} = 10KHz$$



- To achieve good selectivity, bandwidth offered by tuned amplifier should be 10 KHz.
- For tuning of TRF receiver to different station corresponding resonant frequency (f_r) of tuned amplifier has to be changed such that $f_r = f_s$.
- For a tuned amplifier, as the resonant frequency changes, the corresponding bandwidth (BW) also changes. Hence, due to this reason, the selectivity of the TRF receiver will be poor.

  As BW = $\frac{f_r}{Q}$
- If f_r changes then bandwidth also changes, but for keeping the bandwidth same, we have to vary the quality factor (Q) inversely (increase or decrease) with the resonant frequency (f_r) to get a constant bandwidth of 10 KHz.
- This is because selectivity is high bandwidth is equal to 10 KHz.

$$Q = \frac{1}{R}\sqrt{\frac{L}{C}} \left(Q = Quality\ factor\right)$$

$$\& f_r = \frac{1}{2\pi\sqrt{LC}} \left(f_r = resonant\ \ frequency\right)$$

e.g., If

$$f_r = 100\,KHz\ and\ \ Q = 10 \rightarrow BW$$

$$= \frac{100\,KHz}{10} = 10\,KHz$$

If f_r increases to $f_r = 1000\,KHz$

Now for keeping the BW to 10 KHz, Q must be increased.

i.e., BW = $10\,KHz = \frac{1000\,KHz}{Q}$

$Q = 100$

- Thus, for using TRF receiver at good selectivity, we have to manage Q according to f_r (resonant frequency) and this is complex task.
- Hence, the TRF receiver works on poor selectivity and this is the main drawback of the TRF receiver.

Local oscillator:

- Local oscillator is used to generate frequency which is used for down or up-conversion.
- Hartley oscillator is generally used as local oscillator here, and local oscillator frequency is given by f_l

$$f_l = \frac{1}{2\pi\left(L_1 + L_2\right)C}$$

$$f_{l\max} = f_s + IF$$

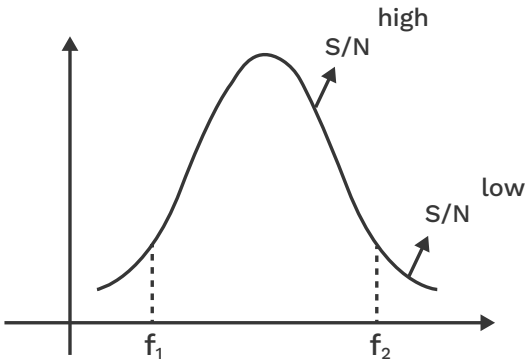
$$f_{l\max} = \frac{1}{2\pi\sqrt{\left(L_1 + L_2\right)C_{\min}}}$$

$$f_{l\min} = f_s - IF$$

$$f_{l\min} = \frac{1}{2\pi\sqrt{\left(L_1 + L_2\right)C_{\max}}}$$

Fidelity:

- It is the ability of receiver to produce all frequency components of transmitted message signal.
- To improve fidelity of receivers, pre-emphasis and de-emphasis (PE and DE) mechanism is used.



- For low frequency components $\frac{S}{N}$ is high but for high frequency components $\frac{S}{N}$ is low, due to this m(t) cannot be recovered properly.
- As S/N of high frequency components is low, therefore for proper recovery of message signal, we use pre-emphasis and de-emphasis.

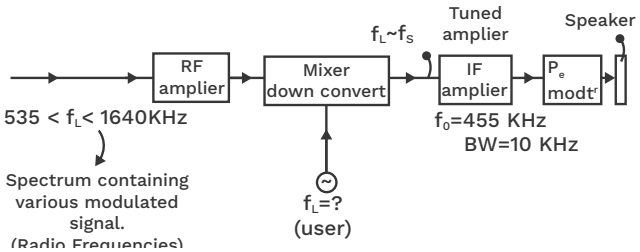
Pre-emphasis:

- It is the process of artificial boosting of high frequency components of original message signal to increases corresponding S/N.
- It is done at transmitter end before modulation of signal.
- High-pass filter is used to amplify high frequency components.

De-emphasis:

- It is reverse of pre-emphasis to get back original message signal.
- It is done at receiver end after demodulation.
- Low-pass filter is used in de-emphasis process.

Superheterodyne Rx:



Let $f_L = 1055\,KHz \rightarrow user$

	F_s	Mixer	If Amplifier	
R1	600 KHz	455 KHz	Yes	Desired station
R2	700 KHz	355 KHz	No	
R3	800 KHz	255 KHz	No	
R4	900 KHz	155 KHz	No	
R5	1510 KHz	455	Yes	Image station

Case 1: $f_{si} > f_s$

$f_s \quad f_l \quad f_{si}$ (If, If)

$f_l - f_s = IF$

$f_{si} - f_s = 2IF$

f_s : *desired freq.*

f_L : *Local oscillator freq*

f_{si} = *freq. of image station*

Case 2: $f_l > f_s$

$f_s < f_l$

$f_s \quad f_l \quad f_{si}$ (If, If)

Case 3: $f_{si} < f_s$

$f_{si} \quad f_l \quad f_s$ (If, If)

$f_s - f_l = IF$

$f_s - f_{si} = 2IF$

$f_l - f_{si} = IF$

Case 4: $f_l < f_s$

$f_{si} \quad f_l \quad f_s$ (If, If)

Example: Superheterodyne Rx: If = 15 MHz

$f_l = 3.5\,GHz$

Frequency of received signal is greater than local oscillator frequency. Then image frequency in MHz = ?

Solution: $f_{si} \quad f_l \quad f_s$ (If, If)

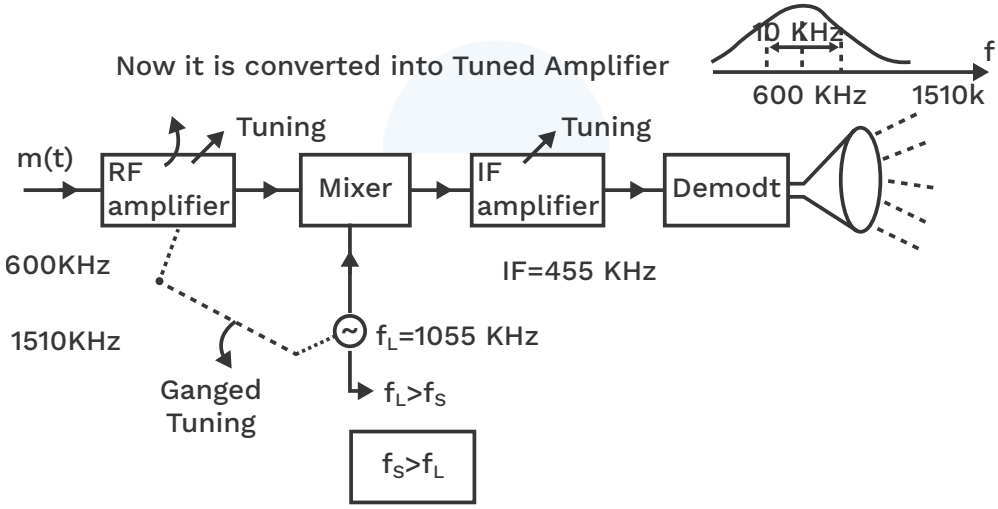
3500 MHz

$f_l - f_{si} = If$

$f_{si} = 3500 - 15$

$f_{si} = 3485\,MHz$

Superheterodyne receivers:



Why $f_l > f_s = ?$

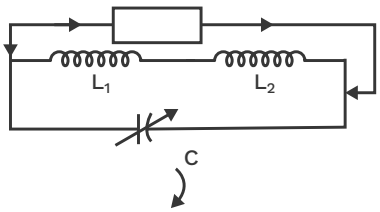
Case 1: $f_l > f_s$ (If, If)

$f_s < f_l < f_{si}$

$f_l = f_s + If \qquad 535\ k < f_s < 1640\ k$

$f_{l\min} = 535k + 455k = 990k$

$f_{l\max} = 1640k + 455k = 2095k$



Local oscillator:

$$f_l = \frac{1}{2\pi\sqrt{(l_1 + l_2)C}}$$

$$f_{l\max} \alpha \frac{1}{\sqrt{C_{\min}}}$$

$$\frac{f_{l\max}}{f_{\min}} = \sqrt{\frac{C_{\max}}{C_{\min}}} = \frac{2095}{990} \approx 2$$

$$\frac{C_{\max}}{C_{\min}} = 4 \rightarrow "Tuning\ Range"$$

Case 2: $f_l < f_s \quad f_{si} < f_l < f_s$

$f_s = f_l + If$

$f_l = f_s - If$

$(f_l)_{\min} = 535k - 455k = 80k$

$(f_l)_{\max} = 1640k - 455k = 1185k$

$$\frac{(f_l)_{\max}}{(f_l)_{\min}} = \sqrt{\frac{C_{\max}}{C_{\min}}}$$

$$Tuning\ Range \leftarrow \frac{C_{\max}}{C_{\min}} = 200$$

Note: '$f_l > f_s$ *is chosen so that tuning range of capacitor remains optimum*'.

Problem Solving Technique

Data: f_s, f_l, IF, f_{si}

Case 1: If relation between f_l *and* f_s is not mentioned:

Assume, $f_l > f_s$

$f_s \quad f_l \quad f_{si}$

1. $f_l = f_s + IF$
2. $f_{si} = f_l + If$
3. $f_{si} = f_l + 2If$

Case 2: When relation between f_l *and* f_s is given:

If: $f_{si} < f_l < f_s$

$\xleftarrow{IF} \xleftarrow{IF}$

1. $f_s = f_l + IF$
2. $f_l = f_{si} + IF$
3. $f_s = f_{si} + 2IF$

If $f_s < f_l < f_{si} \rightarrow case(1)$.

Image refection ratio:

$IRR = \sqrt{1 + P^2Q^2}$

Example: Quality factor of oscillator

$$P = \frac{f_{si}^{\ 2} - f_s^{\ 2}}{f_{si} f_s} \rightarrow "f_{si} > f_s"$$

$P^2Q^2 >>>> 1$

IRR = PQ

Example: A superheterodyne FM receiver operates in the frequency range 88 to 108 MHz. The IF and local oscillator frequencies are chosen such that $f_{IF} < f_{LO}$. The image frequency f_{si} should fall outside of the 88 – 108 MHz range. The minimum value of intermediate frequency f_{IF} required (in MHz) is _____.

Solution:

Let, $f_l > f_s$

$f_{si} = f_s + 2IF$

$$f_{si} < 88\,MHz \rightarrow f_s + 2IF < 88\,MHz \quad (80\,MHz - 108\,MHz)$$
$$\Rightarrow 108\,MHz + 2IF < 88MHz$$

IF < -10 MHz

$$f_{si} > 108\,MHz \rightarrow f_s + 2IF > 108\,MHz \quad (88MHz - 108MHz)$$

IF > 10 MHz

$(If)_{\min} = 10\,MHz$

Let $f_l < f_s$

$f_{si} = f_s - 2IF$

$$f_{si} < 88\,MHz \rightarrow \underset{108MHz}{f_s} - 2IF < 88MHz$$

$IF > 10\,MHz \rightarrow IF_{min} = 10\,MHz$

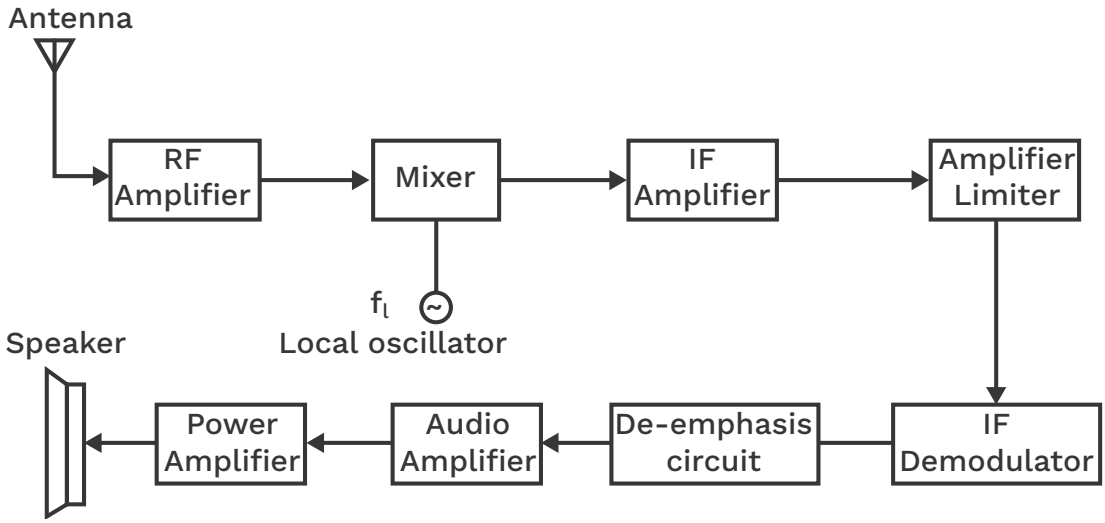
$f_{si} > 108MHz \rightarrow f_s - 2IF > 108MHz$

$IF < -10\,MHz$

Common sense: $f_l > f_s$ $\qquad f_{si} = f_s + 2IF$

FM Receivers

The block diagram of FM receiver is shown below



The block diagram of FM receiver is nearly similar to AM receiver except the two blocks: an amplitude limiter and de-emphasis circuit. They are included before and after the FM demodulator respectively. The operation of all other blocks is same as that of AM receiver.

- In FM modulation, the amplitude of FM wave remains constant and instantaneous frequency of the carrier wave varies linearly with the message signal.
- But, some noise is added with FM wave in the channel during transmission. It may cause variations or fluctuations in amplitude of FM. So, to overcome this fluctuation, amplitude limiter is used.
- As operation of Foster–Seeley demodulator (FSD) is highly sensitive to voltage fluctuations of FM-signals. Hence, we have to use amplitude limiter before FM-demodulator.

Amplitude-limiter:

It is used to remove the variations in amplitude to maintain the amplitude of FM signal constant. De-emphasis network is used after FM demodulator. Now, output of de-emphasis network is passed to the audio amplifier to increase the power level and finally we get the original message signal.

Figure of Merit (FOM):

The ratio of output SNR to input SNR is termed as figure of merit. It is used to characterise the performance of the device, system or method relative to its alternative.

High figure of merit specifies better noise performance of the receiver.

Figure of merit (F) = $\dfrac{S_0 / N_0}{S_i / N_i}$

Noise figure (NF) = $\dfrac{1}{Figure\ of\ merit} = \dfrac{1}{F}$

Noise figure (NF) = $\dfrac{S_i / N_i}{S_0 / N_0}$

Figure of merit for different modulation schemes is as follows:

1. $\left(Figure\ of\ merit\right)_{AM} = \dfrac{\mu^2}{2+\mu^2} = \eta$

The maximum figure of merit is at $\mu = 1$

For sinusoidal signal, maximum figure of merit is 1/3.

For finding FOM of other signals, use respective η.

2. $\left(Figure\ of\ merit\right)_{DSB-SC} = \dfrac{\left(SNR_0\right)_{DSB-SC}}{\left(SNR_i\right)_{DSB-SC}}$

For DSB-SC, $\left(SNR_0\right)_{DSB-SC} = \left(SNR_i\right)_{DSB-SC}$

Hence,

Figure of merit for DSB-SC = 1

3. $\left(Figure\ of\ merit\right)_{SSB-SC} = \dfrac{\left(SNR_0\right)_{SSB-SC}}{\left(SNR_i\right)_{SSB-SC}}$

For SSB-SC,

$\left(SNR_0\right)_{SSB-SC} = \left(SNR_i\right)_{SSB-SC}$

Hence,

Figure of merit for SSB-SC = 1

4. $\left(Figure\ of\ merit\right)_{FM} = \dfrac{\left(SNR_0\right)_{FM}}{\left(SNR_i\right)_{FM}} = \dfrac{3}{2}\beta^2$.

In FM, figure of merit $\alpha\ \beta$

Hence, if β increases, figure of merit increases which leads to increase in quality as $\beta > 1$ for wideband FM (WBFM) and $\beta < 1$ for narrowband FM (NBFM).

Hence, wideband FM is preferred over narrowband FM, because of its high figure of merit.

Example: Calculate the figure of merit for amplitude modulated square wave with $\mu = 0.5$.

Solution:

$\left(Figure\ of\ merit\right)_{AM} = \eta$

$And\ \eta = \dfrac{K_a^2 P_m}{1 + K_a^2 P_m}$

$\left(where\ P_m = Power\ of\ message\ signal\right)$

And for square wave

$P_m = A_m^2$

$\eta = \dfrac{K_a^2 A_m^2}{1 + K_a^2 A_m^2}$

$\eta = \dfrac{\mu^2}{1 + \mu^2} \qquad \left(K_a A_m = \mu\right)$

For $\mu = 0.5$

$\eta = \dfrac{\left(0.5\right)^2}{1 + \left(0.5\right)^2} = \dfrac{1}{5} = 0.2$

$\left(FOM\right)_{AM} = \eta = 0.2$

Hence, figure of merit of amplitude modulated square wave with $\mu = 0.5$ is 0.2.

Exercise

Common data for questions 1 and 2:

Let $m(t) = \cos\left[\left(4\pi \times 10^3\right)t\right]$ be the message signal and $c(t) = 5\cos\left[\left(2\pi \times 10^6\right)t\right]$ be the carrier.

1. c(t) and m(t) are used to generate an AM signal. The modulation index of the generated AM signal is 0.5. Then the quantity $\dfrac{\text{Total sideband power}}{\text{Carrier power}}$ is

(A) $\dfrac{1}{2}$ (B) $\dfrac{1}{4}$

$\dfrac{A_c\mu}{2} \rightarrow LS$ (D) $\dfrac{1}{8}$

Solution: (D)

2. c(t) and m(t) are used to generate an FM signal. If the peak frequency deviation of the generated FM is three times the transmission bandwidth of the AM signal, then the coefficient of the term $\cos[2\pi(1008 \times 10^3 t)]$ in the FM signal (in terms of the Bessel coefficient) is

(A) $5J_4(3)$

(B) $\dfrac{5}{2}J_8(3)$

(C) $\dfrac{5}{2}J_8(4)$

(D) $5J_4(6)$

Solution: (D)

3. A carrier is phase modulated (PM) with frequency deviation of 10 kHz by a single tone frequency of 1 kHz. If the signal tone frequency of 1 kHz. If the single tone frequency of 1 kHz is increased to 2 kHz, assuming that phase deviation remains unchanged, the bandwidth of the PM signal is

(A) $21\,kHz$

(B) $22\,kHz$

(C) $42\ kHz$

(D) $44\ kHz$

Solution: (D)

4. A massage signal $m(t) = A_m \sin(2\pi f_m t)$ is used to modulate the phase of a carrier $A_c \cos(2\pi f_c t)$ to get the modulated signal $y(t) = A_c \cos[2\pi f_c t + m(t)]$. The bandwidth of $y(t)$

(A) Depends on A_m but not on f_m

(B) Depends on f_m but not on A_m

(C) Depends on both $c(t$ and f_m

(D) Does not depends on A_m or f_m

Solution: (C)

Solutions

1. (D)

Given: $m(t) = \cos[(4\pi \times 10^3)t]$

$c(t) = 5\cos[(2\pi \times 10^6)t]$

$m_a = 0.5$

Carrier power is given by,

$$P_c = \frac{A_c^2}{2}$$

Where, A_c = Amplitude of carrier signal

$$P_c = \frac{(5)^2}{2} = 12.5\,\text{W}$$

Total sideband power for signal – tone sinusoidal modulating signal is given by,

$$P_{SB} = \frac{P_c m_a^2}{2} = \frac{12.5(0.5)^2}{2} = \frac{25}{16}\ \text{W}$$

The ratio of total sideband power to carrier power is

$$\frac{P_{SB}}{P_c} = \frac{25}{16 \times 12.5} = \frac{1}{8}$$

Hence, the correct option is (D).

2. (D)

Given: $\Delta f_{\max} = 3(\text{Bandwidth of AM})$

$\Delta f_{\max} = 3(2 f_m)$

$\Delta f_{\max} = 6 f_m$ ...(i)

Expression for WBFM signal is given by,

$$s(t) = A_c \sum_{n=-\infty}^{\infty} J_n(\beta)\cos\left[2\pi(f_c + nf_m)t\right] \quad ....(ii)$$

Ln FM signal, modulation index is given by,

$$\beta = \frac{\Delta f_{\max}}{f_m}$$

From equation (i),

$\beta = 6$

Given term is $\cos[2\pi(1008 \times 10^3)t]$.

On comparing given term with equation (ii),

$\cos[2\pi(f_c + nf_m)t] = \cos[2\pi(1008 \times 10^3)t]$

$f_c + nf_m = 1008 \times 10^3$

$f_c + nf_m = [1000 + 4(2)]10^3$

$10^6 + n(2 \times 10^3) = \left[10^6 + 4(2 \times 10^3)\right]$

On comparing both sides of equation,

$n = 4$

Then the coefficient of the term $\cos[2\pi(1008 \times 10^3)t]$ in the FM signal will be

$A_c J_n(\beta) = 5J_4(6)$

Hence, the correct option is (D)

3. (D)

Given: $\Delta f_{\max} = 10kHz, f_m = 1kHz$

For PM, $\Delta f_{\max} = k_p A_m f_m$

$$\beta = \frac{\Delta f_{max}}{f_m} = k_p A_m$$

The modulation index of PM wave does not depend on modulating frequency. Therefore when modulating frequency is doubled β remains constant.

Modulation index of the PM wave is given by,

$$\beta = \frac{\Delta f_{max}}{f_m} = \frac{10 \times 10^3}{1 \times 10^3} = 10$$

The bandwidth of the PM wave is given by,

$$BW = 2(\beta + 1) f_m$$

$$BW = 2(10+1)2 \times 10^3 = 44\,\text{kHz}$$

Hence, the correct option is (D)

4. (C)

Given: $m(t) = A_m \sin(2\pi f_m t)$ and

$c(t) = A_c \cos(2\pi f_c t)$

$$y(t) = A_c \cos[2\pi f_c t + m(t)]$$

According to Carso's rule, bandwidth of PM signal is given by,

$$BW = 2(\Delta f + f_m)$$

In PM maximum frequency deviation for sinusoidal signal is given by,

$$\Delta f_{max} = k_p A_m f_m$$

Thus, Δf_{max} depends on A_m.

So, bandwidth depends upon both f_m and A_m.
Hence, the correct option is (C).